Terms of Reference templates

Part of D4.3 Guidelines for enabling uptake in project and policy cycles

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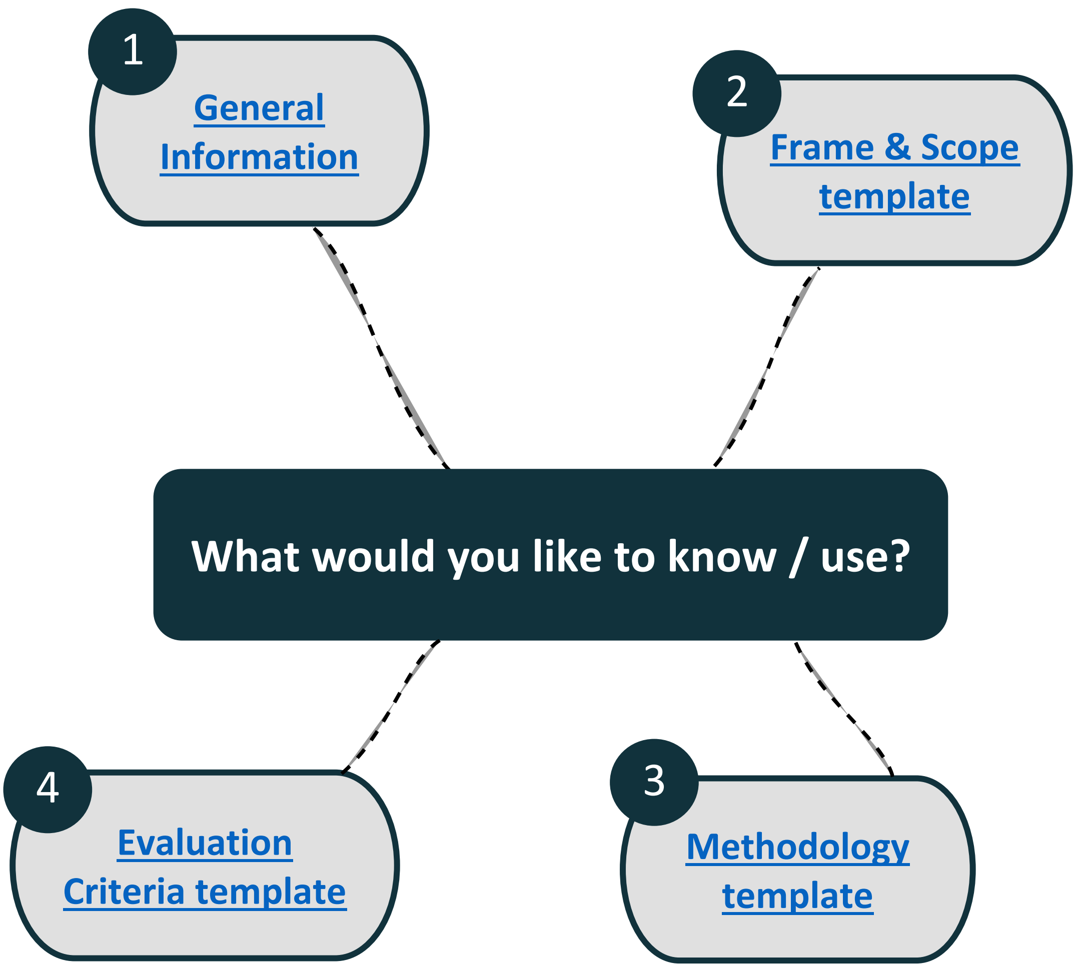
Table of Contents

Welcome to the creation of your Terms of Reference



General information

What would you like to know?

1. [About the guidance]
2. [Recommended workflow: which template and when?]
3. [How to use the templates?]
4. [I need examples of use of the templates!]
5. [Important concepts to know prior to commissioning ecosystem service assessments]

Back to the main landing page

1. How to use the templates?

1.2 Frame & Scope template

1.2.1 Structure of the Frame & Scope section of a Terms of Reference

The guidance in the Frame & Scope templates recommends commissioners to structure their Frame & Scope section of a Terms of reference with four subsections as shown in **Figure 1** below**.**

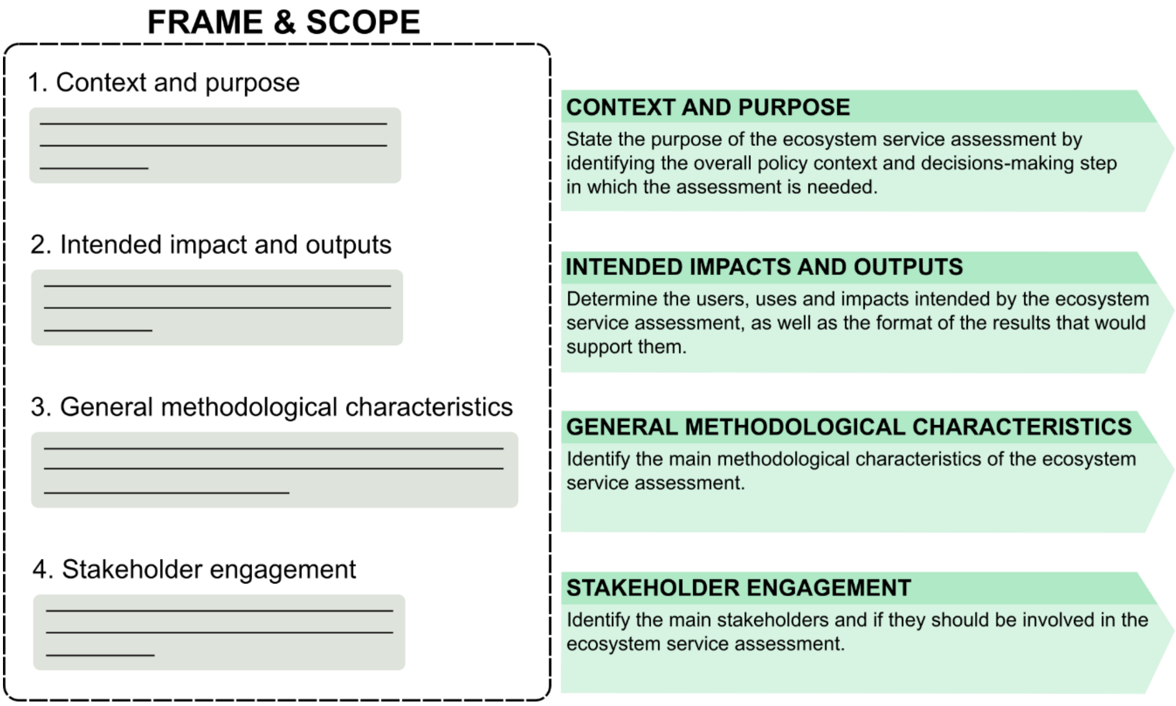


Figure 1 - Recommended structure of the Frame & Scope section of a Terms of Reference to commission an ecosystem service assessment

1.2.2 Writing the Frame & Scope section of a Terms of Reference

Upon arriving on the Frame & Scope, the commissioners will see a list of questions for each of the sections presented in **Figure 1**. Commissioners can choose the questions that they wish to answer before going to the detailed page of the question by following the "Link to the detailed question". On this detailed page for each question, commissioners will have space to answer the question itself, and will have access to both an information box including descriptions and examples, as well as a Terms of Reference box displaying a preformatted text that can directly be used and modified by the commissioners in their own Terms of Reference (**Figure 2**).

The more sections and questions the commissioners cover and answer, the more complete the ecosystem service assessment will be as it will integrate more concepts essential to its design (refer to our website here). The Frame & Scope template is however not exhaustive and if the commissioners wish to add more details to their Terms of Reference, they should modify as needed with the additional support of the advanced content in our report.

Back to the General Information landing page

This box will explain the relevance of the question asked and will give an example of the answer.

This box will display the associated preformatted text for the Terms of Reference. This text can directly be used "as is" or further modified by commissioners to better fit their needs.

Figure 2 - Information associated to each question of all the templates

1.3 Methodology template

1.3.1 Structuring the Methodology section of a Terms of Reference

The guidance in the Methodology templates recommends commissioners to structure their Methodology section of a Terms of reference with four subsections as shown in **Figure 3** below**.**

1.3.2 Writing the Methodology section of a Terms of Reference

Upon arriving on the Methodology template, the commissioners will see a list of questions for each of the sections presented in **Figure 3**. Commissioners can choose the questions that they wish to answer before going to the detailed page of the question by following the "Link to the detailed question". On the question-specific page, commissioners will have space to answer the question itself, and will have access to both an information box including descriptions and examples, as well as a Terms of Reference box displaying a preformatted text that can directly be used and modified by the commissioners in their own Terms of Reference (**Figure 2**).

The more sections and questions the commissioners cover and answer, the more complete the ecosystem service assessment will be as it will integrate more concepts essential to its design (refer to our website here). The Methodology template is however not exhaustive and if the commissioners wish to add more details to their Terms of Reference, they should modify as needed with the additional support of the advanced content in our report.

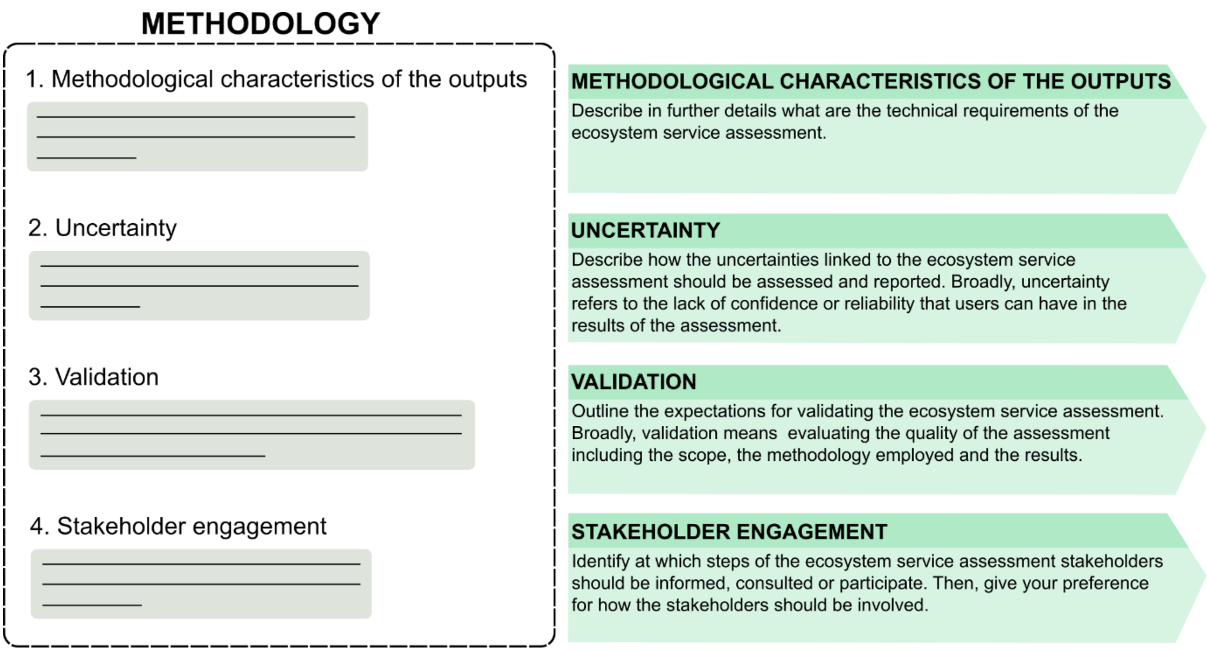


Figure 3 - Recommended structure of the Methodology section of a Terms of Reference to commission an ecosystem service assessment

1.4 Evaluation Criteria template

1.4.1 Structuring the Evaluation Criteria section of a Terms of Reference

The guidance in the Evaluation Criteria templates supports commissioners in structuring their Evaluation criteria section of a Terms of Reference with two subsections as shown in **Figure 4**. **Note that the guidance only provides support to identify evaluation criteria for the scope and the methodology.** There are other evaluation criteria that the commissioners should consider that this guidance document does not advise on such as the budget, or strategy and governance of the commissioners' institution.

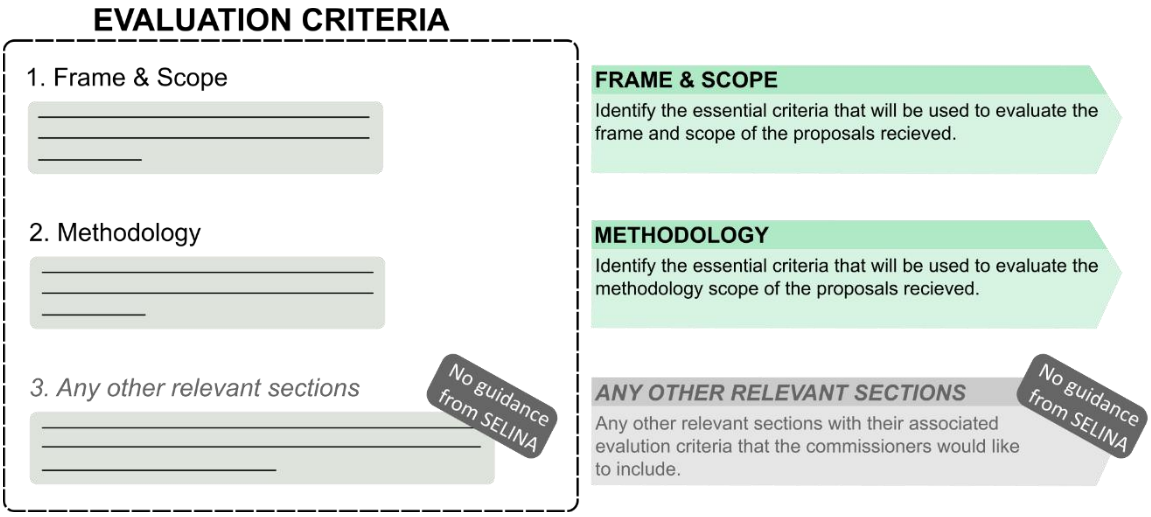


Figure 4 - Recommended structure of the Evaluation Criteria section of a Terms of Reference to commission an ecosystem service assessment

1.4.2 Writing the Evaluation Criteria section of a Terms of Reference

The Evaluation Criteria template differs slightly from the two previous templates. Instead of a list of questions, the commissioners are given a list of evaluation criteria that matches the Frame & Scope and Methodology templates. The commissioners should choose which criteria are essential to them compared with those which could be subject to negotiations with the potential contractors. Commissioners can refer to both the previous templates and further information including descriptions and links to advanced content presented in the information box (**Figure 2**). After choosing the criteria, the commissioners can use the preformatted text displayed in the Terms of reference box to help write the Evaluation Criteria section of their own Terms of Reference (**Figure 2**).

Figure *5* below shows how to use the template form. Writing the Evaluation Criteria section corresponds to Steps 1 to 3.

Back to the General Information landing page

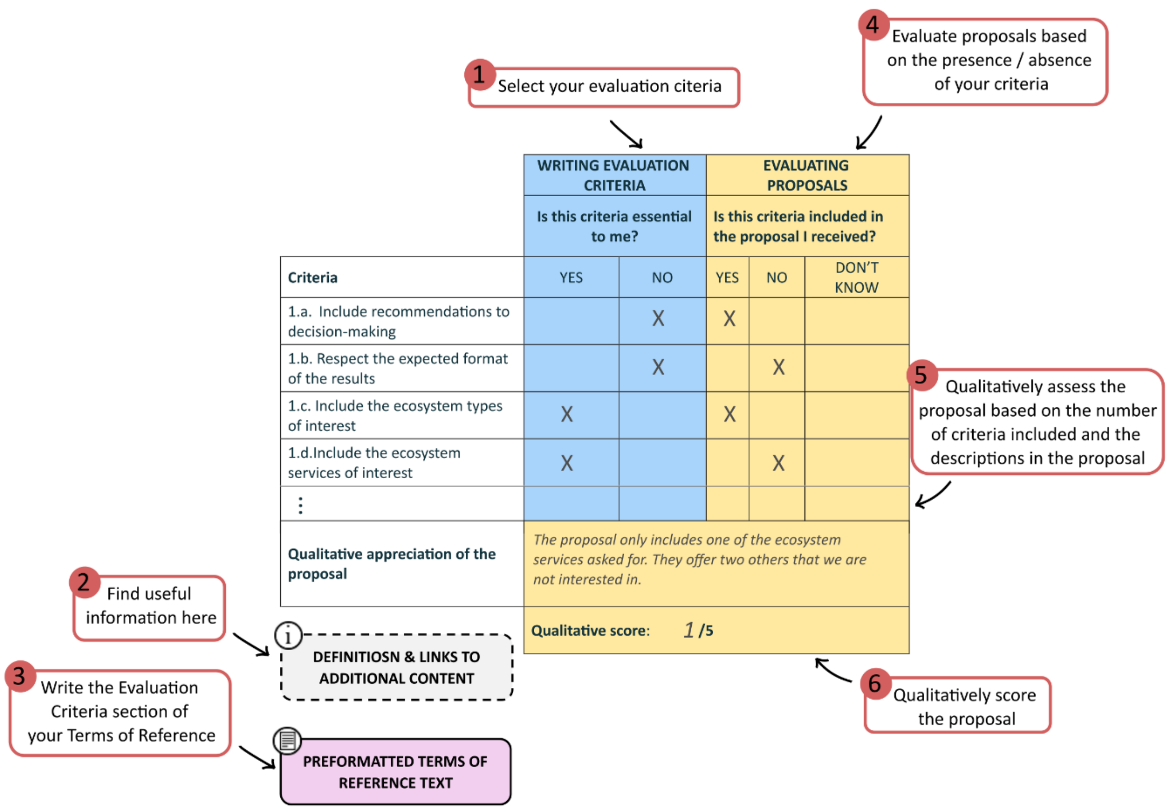


Figure 5 - Using the Evaluation Criteria template to write the Evaluation Criteria section of a Terms of Reference (Steps 1-3) and score and rank proposals (Steps 4-6)

1.4.3 Evaluating proposals with the Evaluation Criteria template

Once the call for tender has ended and the proposals have been received, commissioners can go back to the Evaluation Criteria template to qualitatively score and rank proposals. This phase corresponds to step 4 to 6 in
Figure 5. The scoring is based on:

- the number of criteria that the commissioners has identified as essential and that are included in the proposal received;
- the descriptions of the scope and methodology available in the proposal.

Commissioners can then add the scores given for all sections and rank the proposals. The scoring and ranking are qualitative. The commissioners can use them to decide which proposals should be given further consideration. Commissioners can refer to our advanced content in our report to have further guidance on how to assess the quality of the scope and methodology of a proposal.

2. I need examples of use of the templates!

Examples are provided for each question present in the templates. These examples all refer to the same hypothetical case described in Box 1 below. The main purpose of these examples is to improve the clarity of the questions asked and therefore facilitating a thorough comprehension, by the commissioners, of the concepts embedded in each question.

Box 1- Hypothetical example case

The National Park Mainland is a 5,000 ha large nature area in the east of Belgium. It connects three municipalities and consists mainly of forests, heathland, grasslands, inland dunes and brook valleys. It maintains a rich and rare biodiversity and also offers large recreational opportunities for all ages.

In 2023 it became a National Park. For this nomination, the nature area needed a long-term Masterplan and an Operational plan for the next six years. To assess the progress towards the goals outlined in the Masterplan, an ecosystem assessment was set up, mapping stakeholders, land use, ecosystem condition, and ecosystem services.

In the templates, we answer the questions as if a tender for carrying out this ecosystem assessment was sent out by the managing authority, namely the Agency of Nature and Forest in Flanders. We particularly focus on the ecosystem service assessment. The overall request is to assess a bundle of ecosystem services and to also take into account trade-offs between them.

Back to the General Information landing page

Frame & Scope template

We recommend to read the 5.1 General Information prior to using the template

Below is a list of the questions included in the Frame & Scope template. **Commissioners can use this list to select the questions that are relevant to write the Frame & Scope section of a Terms of Reference.**

Each question is associated with a link to a more detailed page which will provide a box to answer the question, descriptions, examples and the preformatted Terms of Reference text that the commissioners can choose to copy-paste and modify to write the Frame & Scope section of a Terms of Reference to commission ecosystem service assessment.

1. CONTEXT AND PURPOSE

Define the context and aim of the ecosystem service assessment. For this, the commissioners can answer the following questions:

		This question is relevant
1.a. What is(are) the purpose(s) of the ecosystem service assessment? If it is part of a decision-making process, at which step(s) of this process is the assessment needed?	*Link to the detailed question*	YES / NO
1.b. Within this(these) purpose(s) or decision-making step(s), what is the primary objective of the ecosystem service assessment?	*Link to the detailed question*	YES / NO
1.c. Which relevant policies should be considered for the ecosystem service assessment, given the decision-making step and primary objective?	*Link to the detailed question*	YES / NO
1.d. What is the social / demographic context of the ecosystem service assessment?	*Link to the detailed question*	YES / NO
1.e. Should the ecosystem service assessment aim at providing recommendations within the decision-making and policy context?	*Link to the detailed question*	YES / NO

Back to the main landing page

2. INTENDED IMPACTS AND OUTPUTS

Identify and define the intended uses and impacts of the ecosystem service assessment, as well as the format(s) of the deliverables that would support such uses and impacts.

We recommend for a stakeholder analysis to be conducted prior to filling this section as it will help with: (i) identifying the key stakeholders that are most likely to be interested in the outputs, and / or should be involved in the process; (ii) defining the relationships between stakeholders. ***Alternatively, the commissioners can specifically request for such analysis to be conducted by the potential contractor (see 4. STAKEHOLDER ENGAGEMENT section).***

The commissioners can answer the following questions:

2.1. Intended users and uses		**This question is relevant**
2.1.a. How will the commissioner primarily use the result(s) of the ecosystem service assessment?	*Link to the detailed question*	YES / NO
2.1.b. Who is(are) the anticipated secondary user(s) of the results of the ecosystem service assessment?	*Link to the detailed question*	YES / NO
2.1.c. How will secondary users potentially use the result(s) of the ecosystem service assessment?	*Link to the detailed question*	YES / NO
2.2. Intended impacts		**This question is relevant**
2.2.a. What are the main intended impacts of the ecosystem service assessment on decision-making?	*Link to the detailed question*	YES / NO
2.2.b. What are the main intended impacts of the ecosystem service assessment on relevant policies at various spatial scales (*e.g.,* national, regional local)?	*Link to the detailed question*	YES / NO
2.2.c. Are there any other impacts that are intended by the ecosystem service assessment? If yes, which ones?	*Link to the detailed question*	YES / NO
2.3. Formats of the deliverables		**This question is relevant**
2.3.a. What is(are) the expected format(s) for the deliverables of the ecosystem service assessment?	*Link to the detailed question*	YES / NO

Back to the main landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

Define the technical scope of the ecosystem service assessment that is required given the purpose and intended uses and impacts. It is not intended to be exhaustive as a separate Terms of Reference template has been developed to help commissioners identify more in detail what methodology is required for their objective(s). It can be useful to invite relevant stakeholders to scope for the needs of the general methodological characteristics.

If the commissioners feel they do not have sufficient knowledge of which ecosystems / services should be considered, even a general summary of the key habitat or landscape characteristics of the area to be covered by the assessment can be useful to include in the Terms of Reference. In such cases, it also may be helpful to explicitly invite potential contractors to provide some initial thoughts on these questions in their application to the call for tender.

Commissioners can answer the following questions:

3.1. Ecosystem type(s) and service(s)		**This question is relevant**
3.1.a. What ecosystem type(s) is(are) of interest in the ecosystem service assessment?	*Link to the detailed question*	YES / NO
3.1.b. What ecosystem service(s) is(are) of interest in the ecosystem service assessment?	*Link to the detailed question*	YES / NO
3.2. Spatial characteristics		**This question is relevant**
3.2.a. What is(are) geographical location(s) and total area(s) (*i.e.,* spatial extent) of the ecosystem service assessment?	*Link to the detailed question*	YES / NO
3.2.b. At which spatial scale(s) should the results of the ecosystem service assessment be presented (*e.g.,* results aggregated to administrative scales such as regions and counties)?	*Link to the detailed question*	YES / NO
3.3. Temporal characteristics		**This question is relevant**
3.3.a. What is the time period and temporal scale(s) to be considered in the ecosystem service assessment?	*Link to the detailed question*	YES / NO
3.4. Valuation of ecosystem services		**This question is relevant**
3.4.a. Should the ecosystem service(s) be measured in biophysical terms?	*Link to the detailed question*	YES / NO
3.4.b. Should the ecosystem service(s) be measured in terms of their economic value in monetary units?	*Link to the detailed question*	YES / NO
3.4.c. What aspects of ecosystem services should be measured?	*Link to the detailed question*	YES / NO
3.4.d. Should the ecosystem service assessment address social benefits and equity goals?	*Link to the detailed question*	YES / NO

Back to the main landing page

3.4.e. Should the ecosystem service(s) be valued in terms of their relevance to health? If yes, should the assessment look at human, animal, or plant health, or take a One Health approach combining all three with ecosystem health?	*Link to the detailed question*	YES / NO
3.4.f. Should the ecosystem service assessment evaluate the conservation, sustainable supply and / or use of the ecosystem service(s) of interest?	*Link to the detailed question*	YES / NO
3.4.g. Should the ecosystem service assessment include an evaluation of ecosystem condition?	*Link to the detailed question*	YES / NO
3.5. Scenarios		**This question is relevant**
3.5.a. Are future scenarios required in the ecosystem service assessment?	*Link to the detailed question*	YES / NO

4. STAKEHOLDER ENGAGEMENT

Identify and define the desired uses and impacts of the ecosystem service assessment, as well as the format(s) of the deliverables that would support such uses and impacts. Stakeholders are any group, organisation or individual who are interested in and / or might be affected by ecosystem services and any change made to their flows.

Although commissioners might have already carried out an initial stakeholder analysis, we recommend that the commissioners ask potential contractors to conduct a supplementary stakeholder analysis to identify any additional stakeholders that should be involved. For this, the commissioners can answer the following questions:

		This question is relevant
4.a. Should the contractor conduct a stakeholder analysis?	*Link to the detailed template*	YES / NO
4.b. Is it anticipated that any stakeholder(s) will be involved at any stage(s) of the ecosystem assessment process (*e.g.,* assessment development, practice, review, reporting)?	*Link to the detailed template*	YES / NO
4.c. If key stakeholders have already been identified for direct involvement in the ecosystem service assessment: who are they and why have they been selected?	*Link to the detailed template*	YES / NO

Back to the main landing page

Frame & Scope template - detailed questions

1. CONTEXT AND PURPOSE

1.a. What is(are) the purpose(s) of the ecosystem service assessment?
If it is part of a decision-making process, at which step(s) of this process is the assessment needed?

Identify the purpose of the assessment: what will the results of the assessment help you achieve? If part of a decision-making process, identify the step(s) for which you require this assessment: what will the results of this assessment help you achieve in terms of decision-making?

Answer in this box.

Why is this question relevant to me?

Identifying where and why an ecosystem service assessment is needed is the essential first step. It might be that the assessment is needed as part of a decision-process. Depending on where the commissioners are in the decision-making process, the purpose of the ecosystem service assessment will be different: is it to inform a decision? Is it to plan a decision? Is it to monitor the result of a decision?

The overall purpose for conducting an ecosystem service assessment influences the objectives of the assessment, the outputs and its overall process of conducting an assessment including stakeholder engagement. See the decision-making steps for public and private sectors in Annex 8.1 Typology of purposes.

Give me an example answer

The ecosystem assessment is needed for a combination of strategic and operational purposes. It is both demonstrating the importance of a National Park label for nature and society and monitoring that the decisions taken are contributing to the goals set in the Masterplan of the National Park Mainland.

Terms of Reference preformatted text

This is context-dependent, no general text for Terms of Reference is available.

Back to the template landing page

1. CONTEXT AND PURPOSE

1.b. Within this(these) purpose(s) or decision-making step(s), what is the primary objective of the ecosystem service assessment?

Identify and describe the primary objective of the assessment.

Answer in this box.

Why is this question relevant to me?

The primary objective of an ecosystem service assessment will condition the key outputs and methodological needs.

Give me an example answer

The primary objective of the ecosystem assessment is to report on the progress towards the goals of the National Park Mainland. Ecosystem services are also an important part to communicate and value the impact of the National Park label.

Terms of Reference preformatted text

The primary objective of the ecosystem service assessment is [*describe your objective*].

Back to the template landing page

1. CONTEXT AND PURPOSE

1.c. Which relevant policies should be considered for the ecosystem service assessment given the decision-making step and primary objective?

Identify the international and national policies that should be considered in the ecosystem service assessment.

Answer in this box.

Why is this question relevant to me?

Once the decision-making step and the primary objective have been identified, it is important to identify the policies that should be considered during this assessment. This ensures that the assessment and its outputs are relevant, actionable and aligned with the current policies.

Give me an example answer

Two policies are of relevance in our context:

- The nature area is selected as National Park with subsidies from the Flemish government. 6-yearly reporting of the status to the National Park Bureau is obliged. It is important that the assessment looks at criteria that needs to be in the Status report.
- Flemish biodiversity plan.

Terms of Reference preformatted text

This is context-dependent, no general text for Terms of Reference is available.

Back to the template landing page

1. CONTEXT AND PURPOSE

1.d. What is the social / demographic context of the ecosystem service assessment?

Identify the key social and demographic characteristics of the area and / or populations interacting with the ecosystems to be assessed. These characteristics may include age distribution, cultural and ethnic composition, income levels, education, livelihood types etc.

Answer in this box.

Why is this question relevant to me?

The ecosystem service assessment may consider the social and demographic context to ensure that it reflects the needs and realities of the people and communities interacting with the ecosystems. Doing so can reveal social or economic challenges, highlight vulnerable or priority groups, and identify population trends that influence the relevance and use of the assessment's outputs. This information further supports tailored communication, inclusive engagement, and more effective, equitable reporting.

Give me an example answer

Recreational development and industrialisation led to prosperity in the region. The social and demographic context of the project is quite broad. The National Park Mainland lays in the perimeter of three municipalities with a wide range of age groups, economic status and ethnic composition. Furthermore, the area attracts visitors from all over the country.

Terms of Reference preformatted text

This is context-dependent, no general text for Terms of Reference is available.

Back to the template landing page

1. CONTEXT AND PURPOSE

1.e. Should the ecosystem service assessment aim at providing recommendations within the decision-making and policy context?

Answer yes, no, maybe, or don't know. Providing recommendations means identifying and describing actionable suggestions to improve or bridge an existing gap of a strategy or situation.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

The need for recommendations influences a number of aspects in an ecosystem service assessment around the engagement of stakeholders.

Give me an example answer

Maybe. The purpose of the assessment is reporting but based on the results of the assessment, recommendations could be derived for the development of the next Operational plan.

Terms of Reference preformatted text

Based on the results, the ecosystem service assessment might aim at providing recommendations on [*list the decision-making context where recommendations are needed*]

Back to the template landing page

2. INTENDED IMPACTS AND OUTPUTS

2.1. Intended users and uses

2.1.a. How will the commissioner primarily use the result(s) of the ecosystem service assessment?

Identify the uses that the results of the ecosystem service assessment will have. What use(s) justifies the sponsoring of the call for tender?

Answer in this box.

i

Why is this question relevant to me?

The expected primary uses of the assessment will condition the types and format of the outputs needed.

Give me an example answer

The results will be primarily used to report to the National Park Bureau on the progress and development of the National Park Mainland. In addition, the results will be used to show the importance of the National Park label for local economy and well-being of the surrounding population.

Terms of Reference preformatted text

The results of this assessment will be primarily used by the commissioner to/for [*list the expected uses that the commissioner will have from the outputs generated by the assessment*].

Back to the template landing page

2. INTENDED IMPACTS AND OUTPUTS

2.1. Intended users and uses

2.1.b. Who is(are) the anticipated secondary user(s) of the results of the ecosystem service assessment?

Identify the additional stakeholders who will be using the outputs of the assessment. By "secondary user(s)" we mean stakeholder(s) who might be interested in directly and actively utilising the outputs of the assessment.

Answer in this box.

Why is this question relevant to me?

It is important to not restrict to the users to the commissioner only as the results of the assessment could be useful for other parties. Identifying these secondary users allows to have stronger insights on the impacts of assessment and what types and format of outputs are needed.

Give me an example answer

Other users of the results could be members of the municipalities surrounding National Park Mainland, nature organisations and other National Parks in Belgium.

Terms of Reference preformatted text

Results should also be made available and usable by [*list the secondary users*]. It is anticipated that these parties will primarily use the direct outputs of the assessment as [*include the answer to question 2.1.c presented on the next page*].

Back to the template landing page

2. INTENDED IMPACTS AND OUTPUTS

2.1. Intended users and uses

2.1.c. How will secondary users potentially use the result(s) of the ecosystem service assessment?

Identify the uses that the outputs of the ecosystem service assessment will have for other stakeholders than the commissioner of the assessment.

Answer in this box.

Why is this question relevant to me?

The expected secondary uses of the assessment will condition the types and format of the outputs needed.

Give me an example answer

Secondary users could potentially use the results to:

- communicate that nature not only has biodiversity benefits but also social and economic benefits.
- put these socio-economic benefits on the map to support municipal budgets

The approach and methodology may be used by other National Parks in Belgium.

Terms of Reference preformatted text

Results should also be made available and usable by [*include answer to question 2.1.b on the previous page*]. It is anticipated that these parties will primarily use the direct outputs of the assessment as [*list the uses expected from the secondary users*].

Back to the template landing page

2. INTENDED IMPACTS AND OUTPUTS

2.2. Intended impacts

2.2.a. What are the main intended impacts of the ecosystem service assessment on decision-making?

Identify and describe the main intended impacts of the assessment on the decision-making. "Impacts" refer to any positive or negative medium and long term effects or influences that the results of an ecosystem service assessment, for example when used to inform policy or design practical interventions, will have upon a decision-making context, society, and / or help to address societal challenges.

Answer in this box.

Why is this question relevant to me?

Considering the intended impacts while asking for an ecosystem service assessment is part of making the results of the assessment actionable.

Give me an example answer

The approach and results of the ecosystem assessment will help the National Park Mainland to show its progress towards the goals of the Masterplan and report this to the National Park Bureau in order to receive financial support.

The results of the assessments will inform policy decisions on conservation measures.

Terms of Reference preformatted text

Outputs of the ecosystem service assessment are intended to directly impact [*list the primary impacts identified*].

Back to the template landing page

2. INTENDED IMPACTS AND OUTPUTS

2.2. Intended impacts

2.2.b. What are the main intended impacts of the ecosystem service assessment on relevant policies at various spatial scales (*e.g.,* national, regional, local)?

Identify and describe the main intended impacts on policy formulation or implementation due to the results of the assessment, assuming they are used by the other stakeholders. "Impacts" refer to any positive or negative medium and long term effects or influences that the results of an ecosystem service assessment, for example when used to inform policy or design practical interventions, will have upon a decision-making context, society, and / or help to address societal challenges.

Answer in this box.

Why is this question relevant to me?

Considering the impacts on policies at various spatial scales when requesting an ecosystem assessment is part of adopting a broader transformative approach. This perspective aids in identifying leverage points that the results may have on society and governance systems at local, regional, national, and global levels.

Give me an example answer

The results will inform the national government on the progress the National Park Mainland and also may also generates input to the European obligations of the Nature Restoration Law.

The results of the ecosystem assessment will underline the importance of nature into the socio-economic fabric of the municipalities and will inform local budget and policy decisions on conservation measures.

Terms of Reference preformatted text

The outputs of the ecosystem service assessment are also intended to bring insights to [*list the policies or policy-makers that the outputs will directly impact*] by [*describe the intended impact on the policies*].

Back to the template landing page

2. INTENDED IMPACTS AND OUTPUTS

2.2. Intended impacts

2.2.c. Are there any other impacts that are intended by the ecosystem service assessment? If yes, which ones?

Identify and describe any other intended impacts that the results of the assessment are expected to have if they are used by stakeholders. "Impacts" refer to any positive or negative medium and long term effects or influences that the results of an ecosystem service assessment, for example when used to inform policy or design practical interventions, will have upon a decision-making context, society, and / or help to address societal challenges.

Answer in this box.

Why is this question relevant to me?

The ecosystem service assessment might have impacts beyond the expected scope of the decision-making within which the assessment has been commissioned. Identifying and understanding these supplementary impacts are essential to have the full area of influence of the assessment. It will also serve to the completion of a risk assessment and management plan.

Give me an example answer

The ecosystem service assessment will enhance the positive attitude towards the National Park label; as not all stakeholders are persuaded that it will have a positive impact in the region.

Terms of Reference preformatted text

Beyond the direct impacts mentioned above, the outcomes of this assessment should also act on/influence/affect [*list who or what will be affected by the outputs of the assessment*] through/by [*describe the expected impact*].

Back to the template landing page

2. EXPECTED IMPACTS AND OUTPUTS

2.3. Formats of the deliverables

2.3.a. What is(are) the expected format(s) for the deliverables of the ecosystem service assessment?

Select all the formats that apply.

Possible formats	Why should choosing this format?	Results should be presented like this
Executive summary	Executive summary can be relevant if the key users are non-expert and need to grasp quickly the essential findings and implications without going through the technical details. They are very useful tools to ensure that key messages are communicated clearly and effectively.	YES / NO / MAYBE / DON'T KNOW
Technical report(s)	Technical reports are important tools for users requiring detailed data-driven insights, or if the methodology is a key result of the assessment.	YES / NO / MAYBE / DON'T KNOW
Presentation / Briefing	Presentations and briefings are very user-friendly and allow to have an overview of the results by the contractors themselves. They are the opportunity for the commissioners to ask questions and clarification with the contractor, ensuring that they understand the work and that the fundings will meet their needs.	YES / NO / MAYBE / DON'T KNOW
Training material	Requesting training material should be particularly considered when there is an objective of capacity building from users. Training can be both made for trainers and trainees.	YES / NO / MAYBE / DON'T KNOW
Tool(s)	Asking for the development of a special tool particularly useful if the results of the assessment should be operationalised so the users can reproduce the analyses with different data or over time. A tool ensures accessibility, standardizsation, replicability and reproducibility.	YES / NO / MAYBE / DON'T KNOW
Scientific publication	Publishing in the scientific community allows to reach a wider audience and contribute to the development of knowledge in ecosystem services. It can be relevant to aim for a scientific publication to provide more credibility and authority to the results of the ecosystem service assessment.	YES / NO / MAYBE / DON'T KNOW
Maps	Maps are ideal deliverables to see spatial patterns, trends and relationships. They are also very intuitive and easily understandable by a wide range of users.	YES / NO / MAYBE / DON'T KNOW
Tables	Tables allow to easily summarise and structure data and results. In some contexts such as looking at the provision of ecosystem services across time, or ecosystem accounting, they are a key format to request.	YES / NO / MAYBE / DON'T KNOW
Videos	Videos are powerful tools especially when it comes to communicating the results to a broader audience. They are easy to understand and can more easily popularize complex concepts through storytelling. They are also very useful to represent dynamic environmental/ecological processes and changes.	YES / NO / MAYBE / DON'T KNOW
Other (specify)		YES / NO / MAYBE / DON'T KNOW

Why is this question relevant to me?
The formats of the deliverables impact the methodology that will be applied in the ecosystem service assessment including data inputs and analyses. The choice of the format will also strongly affect the expected impacts described previously as they condition the accessibility, usability, and actionability of the results.

Give me an example answer
The ecosystem service assessment will be delivered in the following formats:

- An executive summary informing policy makers;
- A technical report so that the approach can be replicated in the future and by other National Parks;
- Tables to easily summarise the results of the assessment;
- A lay man fact sheet/infographic to use in the communication towards citizens and other stakeholder groups.

Terms of Reference preformatted text
The specific deliverables required in this assessment are: [*list the formats selected above*]

- An executive summary of [*if not applicable to all results, precise the results for which this format applies*] designed for [*precise the target audience if relevant*]
- Technical report covering [*if not applicable to all results, describe the results for which this format applies*] designed for [*precise the target audience if relevant*]
- Presentation and / or briefing [*if not applicable to all results, precise the results for which this format applies*] designed for [*precise the target audience if relevant*]
- A (series) of training material designed for [*precise the target audience*]. This training would have to be run [*if relevant, precise the number of times per year/month*]
- A tool designed at [*goal of the tool*]. The format of the tool should be aligned with its intended uses
- Maps of [*if specific maps are required, describe what the maps should display*]
- Tables of [*if specific tables are required, describe what the maps should display*]

- A short videoclip of [*if not applicable to all results, precise the results for which this format applies*] designed for [*precise the target audience if relevant*]
- Scientific publications
- [*describe any other required formats*]

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.1. Ecosystem type(s) and service(s)

3.1.a. What ecosystem type(s) is(are) of interest in the ecosystem service assessment?

Identify and define the ecosystem type(s) that should be focused on in the assessment. Ecosystem types refer to different environments where living things interact with each other and their surroundings. Examples of ecosystem types include forests, city parks, wetlands, sand dunes etc.

Answer in this box.

Why is this question relevant to me?

An ecosystem service assessment aims at evaluating the current status of ecosystem service(s) in a specific ecosystem types and the change(s) in provision of this service over space and time. To do so, the ecosystem types providing the ecosystem services of interest should be identified and defined. A number of standard classifications exists. If the purpose of the assessment is restricted to national uses and impacts, national classifications can be preferred. International classifications can also be used such as European Nature Information System, International Union for Conservation of Nature Global Ecosystem Typology or the System of Environmental Economic Accounting – Ecosystem Accounting ecosystem type reference classification.

If the commissioners are not clear on which ecosystem types should be focused on, a brief description of the area of interest is sufficient. Alternatively, it might also be considered to refer to experts to identify and define them, or leave the call of tender open for suggestions from potential contractors.

Give me an example answer

Ecosystem types are all types found within National Park Mainland:

- Forests
- Grasslands
- Heathland
- Inland wetlands
- Sparsely vegetated areas/land dunes

For the condition assessment only forests will be assessed.

Back to the template landing page

Terms of Reference preformatted text

Ecosystem types of interest in this ecosystem service assessment:

- [*list the ecosystem types of interest*]

Their definitions can be found [*link to the classification system used or other official guidance*].

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.1. Ecosystem type(s) and service(s)

3.1.b. What ecosystem service(s) is(are) of interest in the ecosystem service assessment?

Identify and define the ecosystem service(s) that should be focused on in the assessment. Ecosystem services are the contributions of ecosystems that support economic activities and other aspects of human well-being, such as urban cooling, flood protection, pollination, timber and food provision etc.

Answer in this box.

Why is this question relevant to me?

An ecosystem service assessment aims at evaluating the current status of ecosystem service(s) in a specific ecosystem types and the change(s) in provision of this service over space and time. To do so, the ecosystem services of interest should be identified and defined. As for ecosystem types, a number of standard classifications exist. Such as Classification of Ecosystem Services (CICES), the System of Environmental Economic Accounting – Ecosystem Accounting (SEEA-EA), the National Ecosystem Services Classification System (NESCS Plus), The Economics of Ecosystems and Biodiversity (TEEB), the Millenium Ecosystem Assessment (MEA), or the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). If the commissioners wish to use their own definition, it is recommended to underline the reasons behind this change from more official definitions. The differences between the official and in-house definition should be clearly highlighted.

If the commissioners are not clear on which ecosystem services should be focused on or how to define them, it might be considered to refer to experts. Alternatively, the call of tender can be left open for suggestions from potential contractors on that aspect.

Give me an example answer

Inherent to the ecosystem service approach it is best to assess the total bundle of ecosystem services relevant for National Park Mainland. The relevance will be defined together with different stakeholders in the area.

Terms of Reference preformatted text

The ecosystem services that should be evaluated in this assessment are:

- [*list the ecosystem services of interest*]

Their definition can be found [*link to the classification system used, other official guidance, or commissioners' own definition*].

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.2. Spatial characteristics

3.2.a. What is(are) the geographical location(s) and total area(s) (*i.e.,* spatial extent) of the ecosystem service assessment?

Identify the geographical location(s) and total are(s) in which the assessment should be conducted. The geographical location and the total area (e.g., continental, national, regional, local) form the spatial extent.

Answer in this box.

Why is this question relevant to me?

These are basic information of an ecosystems service assessment. The geographical location and spatial extent of the assessment will influence the data to be used and the overall methodological approaches.

When defining the geographical location and the spatial extent within which the assessment will be carried out, it is important to be aware of possible misalignments between the geographical areas and spatial extent at which the ecosystem services are provided and the ones at which people benefit from them. Benefits from ecosystem services can sometimes occur outside of the geographical area and spatial extent at which the ecosystem services are provided. For example, a forest located in the upper part of a mountain provides the soil retention service. The benefit from this service is the reduction in land-sliding. The beneficiaries of this service are the residents living outside of the forest, at the bottom of the mountain.

If the assessment is carried out at several spatial scales, this can support the establishment of a decision-making or management that will vary with scales to ensure its suitability and relevance.

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on spatial and temporal characteristics!

Give me an example answer

The assessment takes place on the geographical location and local scale of the National Park Mainland. If possible regional scale for ecosystems such as flood protection.

Terms of Reference preformatted text

The ecosystem service assessment is to be conducted in [*geographical location, if relevant add the geographical coordinates*] at [*spatial extent(s)*] level.

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.2. Spatial characteristics

3.2.b. At which spatial scale(s) should the results of the ecosystem service assessment be presented (*e.g.,* results aggregated to administrative scales such as regions and counties)?

Select what applies. The spatial scales apply within the geographical location and spatial extent chosen of the ecosystem service assessment.

Possible level of spatial representation	Description	Results should be presented like this
Fully spatially explicit	A fully spatially explicit ecosystem service assessment does not aggregate the results. It displays ecosystem types and services with fine-grained spatial detail. This is useful to capture fine-scale spatial patterns and interactions that influence ecosystem services, and to inform site-specific management actions or interventions. Use when you need detailed analyses that account for spatial heterogeneity and variability. Careful, it usually requires more time and resources and data availability may be a limiting factor.	YES / NO / MAYBE / DON'T KNOW
Aggregation at ecological scale	In this option, results aggregate ecosystem types and services by groups that are ecologically meaningful, presenting one averaged figure. For example, forests can be grouped into deciduous and coniferous forests. This is useful to understand the supply of ecosystem services across natural boundaries (e.g., carbon sequestration in a forest ecosystem depends among others on the species composition). Use when the natural boundaries of the ecosystem processes are more relevant than administrative divisions. Particularly useful for ecosystem service dependent on ecological units, such as hydrological services, nutrient cycling, or habitat connectivity.	YES / NO / MAYBE / DON'T KNOW If YES, precise the ecological scale:
Aggregation at administrative scale	In this option, results for ecosystem types and services are aggregated by administrative units (e.g., municipality, county), presenting one averaged figure. This is useful to align ecosystem service assessments with decision-making frameworks and administrative jurisdictions that manage resources and implement policies. Use when the focus is on governance, policy-making, or management actions that are implemented at administrative levels or on integrating ecosystem service assessments into planning and policy processes.	YES / NO / MAYBE / DON'T KNOW If YES, precise the administrative scale:

Why is this question relevant to me?

The spatial distribution of ecosystem services supply and use, but also the benefits from ecosystem services are highly variable, hence this is a crucial aspect to consider in decision-making. Identifying discrepancies in space is important to better manage inequalities (e.g., in distribution of or access to ecosystem services), but also manage synergies, interdependencies and trade-offs between different ecosystem services and / or their associated benefits.

As for the definition of the geographical location and spatial extent, when defining the spatial scale(s) at which the assessment will be carried out, it is important to be aware of possible misalignments between the spatial scales at which the ecosystem services are provided and the ones at which people benefit from them. Benefits from ecosystem services can sometimes occur at spatial scales at which the ecosystem services are provided.

Sometimes, a multi-scale approach may be necessary to address different aspects of ecosystem supply and use. For example, ecological and administrative spatial scales can be used in conjunction if the decision-making context is directed at specific ecosystem types within certain administrative boundaries.

For decision-making purposes or computation reasons, a fully spatially explicit assessment may not be relevant. If the decision-making only requires county or municipal values to be informed or decided on, aggregating the results of the assessment to these units is sufficient. Also, fully spatially explicit analyses can be resource-intensive in terms of data acquisition, processing, and analysis. Using aggregated data where high resolution is unnecessary can save time and resources while still providing useful insights.

Considering the (appropriate) level of spatial representation ensures that an ecosystem service assessment is both precise and relevant to the contexts and scales at which management and policy decisions are made. The disaggregation through the spatial scales chosen should be restricted to the geographical location and spatial extent chosen.

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on spatial and temporal characteristics!

Give me an example answer

The assessment needs to be as spatially explicit as possible within the context of the available data. If not possible, at the least aggregation per biotope is needed.

Terms of Reference preformatted text

Within the geographical and spatial extent of this ecosystem service assessment, the results should be presented as [*fully spatially explicit; aggregated at precise the ecological or administrative scale(s) of relevance*]

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.3. Temporal characteristics

3.3.a. What is the time period and temporal scale(s) to be considered in the ecosystem service assessment?

Identify the total time period (or interval) for which the ecosystem service assessment is to be conducted. It may also be a single (e.g., annual) assessment without repetition. Then specify the temporal scale, in which you need the results reported. The temporal scale represents the unit of time in which the results should be presented (aggregated if needed) such as by year, by quarter, by month, etc.

Answer in this box.

Why is this question relevant to me?

As ecosystem services (supply, use, and demand) and their benefits vary across time, it is important to clearly indicate the period of time the ecosystem service assessment needs to cover (*e.g.,* 2020-2025). In a second step, it is also necessary to specify the temporal scale in which the results should be reported such as annual average, quarterly or monthly averages. An overall example would be an assessment conducted for the time period 2020-2025 with results broken down by quarters.

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on spatial and temporal characteristics!

Give me an example answer

The assessment should be conducted over the years 2024-2030. Results should be reported for each year within this time period.

Terms of Reference preformatted text

The ecosystem service assessment should be conducted for [*indicate the period of time*]. The results should be reported [*temporal scale*].

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.4. Valuation of ecosystem services

3.4.a. Should the ecosystem service(s) be measured in biophysical terms?

Answer yes no, maybe, or don't know. Valuing, or quantifying, ecosystem services in biophysical terms means that the assessment measures the physical quantity of the ecosystem service (e.g., tonnes of crops produced, tonnes of carbon sequestrated, number of recreational visitors).

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Biophysical valuation enables us to assess the physical/material contribution of ecosystems to human well-being as well as helps to identify how ecosystem characteristics affect the level of ecosystem services. Biophysical ecosystem service assessments can also enable us to evaluate whether the levels of use of ecosystem services correspond with those of the ecosystem's capacity to supply services.

Give me an example answer

Yes, ecosystem services need to be measured in biophysical terms.

Terms of Reference preformatted text

The ecosystem service assessment should measure ecosystem service in biophysical terms.

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.4. Valuation of ecosystem services

3.4.b. Should the ecosystem service(s) be measured in terms of their economic value in monetary units?

Answer yes, no, maybe, or don't know. Economic value is a measure of the contribution of the ecosystem service(s) to human wellbeing. It is generally measured in monetary units to facilitate direct comparison and aggregation with the values of other goods and services (e.g., for inclusion in decision support frameworks such as cost-benefit analysis). It is recommended to first value service in biophysical terms before valuing them in monetary terms.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Economic value in monetary terms facilitates a more direct link to the economy and provides a measure of the importance of ecosystem services to human wellbeing.

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on economic valuation!

Give me an example answer

Yes, where possible the economic value of ecosystem services should be quantified in monetary units.

Terms of Reference preformatted text

The ecosystem service assessment should measure the economic value of ecosystem service in monetary terms.

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.4. Valuation of ecosystem services

3.4.c. What aspects of ecosystem services should be measured?

Select what applies.

Aspects of ecosystem services that can be measured	Description	This aspect should be measured
Ecosystem service supply	The supply of ecosystem services is the measure in biophysical units of the contribution of ecosystems to the economy or society more generally (*i.e.,* what quantity is being supplied?). It can be further valued in terms of other values such as their economic value in monetary units (*i.e.,* how much does this supply represent in the economy?). We recommend the supply to be assessed and reported together with the use of the ecosystem service.	YES / NO / MAYBE / DON'T KNOW
Ecosystem service use	The use of ecosystem services refers to the measure in biophysical units of the utilisation or consumption of the service by users or those who benefit from the services (*i.e.,* what quantity is being used?). It can be further valued in terms of other values such as their economic value in monetary units (*i.e.,* how much does this use represent in the economy?). We recommend the use to be assessed and reported together with the supply.	YES / NO / MAYBE / DON'T KNOW
Ecosystem service demand	The demand for ecosystem services is the measure in biophysical units of the need for specific ecosystem service(s) by society, particular stakeholder groups or individuals (*i.e.,* what quantity of an ecosystem service is needed?). For instance, the area of pollination dependent crops is a way of assessing the need for pollination services by farmers. It can be further valued in terms of their economic value in monetary units (*i.e.,* how much does this need represent in the economy?). The demand may not be equal to the use, as needing ecosystem services doesn't mean that they are supplied, accessed and used. The assessment of ecosystem service(s) demand can be of interest in scenario analyses or to identify areas with deficits of ecosystem service(s).	YES / NO / MAYBE / DON'T KNOW

i

<u>**Why is this question relevant to me?**</u>

An ecosystem service assessment aims at evaluating the current status of ecosystem service(s) in a specific ecosystem types and the change(s) in supply, use and / or demand of this service over space and time. To do so, the *supply* (*i.e.,* provision) of the ecosystem services of interest should be quantified whether it is in biophysical and / or economic values. The *supply* is the basic first information that should be reported in an ecosystem service assessment to be relevant for decision-making.

However, focusing only on the *supply* does not inform on who is consuming the ecosystem services, neither on how the quantity of services is being used by the consumers / users. We recommend the *supply* to be assessed and reported together with the *use* to understand the full flow of ecosystem services: how much is being produced? And who is using and how much is being used? Identifying who is using / benefitting from the ecosystem services and how much of the provision of these services is being consumed by users / beneficiaries is central to any decision-making. This allows to identify who will be impacted by a change in provision of the services and how they will be affected.

Finally, assessing the demand for ecosystem services helps estimating whether the provision of services is sufficient to meet their societal demand. For instance, are there enough pollinators to fulfil the needs of pollination dependent crops?

It is important to note that the demand does not equal the use. In the example above, we can imagine that some farmers need crop-pollination to produce food but don't have the necessary ecosystem types providing this service around their crops. This means that these farmers need pollination but don't have access to this service and use it. Inversely, some other farmers may have their crops surrounded by ecosystem types providing the pollination service but only grow crops that are non-pollinator dependent. This means that these farmers could use the pollination service but they don't need it.

Where can I find more questions to be more detailed in my Terms of Reference?
Visit our advanced content on sustainability and condition!

Give me an example answer
Supply and use should be measured. The demand is not necessarily requested but could be interesting to know for certain ecosystem services such as pollination and recreation in order to detect deficits.

Terms of Reference preformatted text
The ecosystem service assessment should measure the [*supply, use, demand*] of the ecosystem service(s) of interest.

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.4. Valuation of ecosystem services

3.4.d. Should the ecosystem service assessment address social benefits and equity goals?

Answer yes, no, maybe, or don't know. Social benefits are improvements in people's quality of life. Equity goals focus on guaranteeing that these are fairly distributed among social groups. Social benefit is a measure of the contribution of ecosystem services to societal well-being and equity. It can be quantified using metrics such as availability and access to ecosystem services, reduction in social vulnerabilities, social inclusion, etc.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Including social and equity metrics ensures that the assessment addresses inequalities, particularly for vulnerable or underserved populations.

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on social benefits!

Give me an example answer

Yes, if information is available on social benefits, such as nature contact and nature connectedness (*i.e.,* the way people connect to nature and social cohesion), need to be taken into account.

Terms of Reference preformatted text

The ecosystem service assessment should measure the social value of ecosystem services using relevant metrics.

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.4. Valuation of ecosystem services

3.4.e. Should the ecosystem services be valued in terms of their relevance to health? If yes, should the assessment look at human, animal, or plant health, or take a One Health approach combining all three with ecosystem health?

Answer yes, no, maybe, or don't know. Ecosystem services are crucial to health, including the health dimension in an ecosystem service assessment allows to take into account the wider impacts of changes in ecosystem services on the society and the environment. The health scope will inform the selection of assessment methodology, stakeholder engagement and reporting. The One Health approach is an approach that recognises the interconnection between people, animals, plants and their shared environment.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

This ecosystem service assessment may consider the importance of ecosystem services to:

- health benefits, health risks and / or health outcomes,
- relationships between ecosystems, their services and specific priority health issues,
- delivery of health services or planning of health interventions,
- economic importance to private sector health entities,
- the benefits of the assessment to health sciences, etc.

It is important to identify the specific dimensions of health to be included, and consider whether established approaches such as One Health may be useful; this will determine the kinds of data, metrics, indicators and stakeholders to be incorporated into the assessment process. Interlinkages between human health and the health of other biota and of ecosystems are well established and form the basis of the One Health approach; an assessment considering any one compartment in isolation may not provide a sufficiently robust output for some intended uses or users, so this question requires careful consideration

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on health benefits!

Give me an example answer

Yes. Impacts on human health should be taken into account qualitatively and where possible quantitatively. Ecosystem health is embedded in the ecosystem condition assessment requested in this tender.

Terms of Reference preformatted text

The ecosystem service assessment should measure the value of ecosystem services in terms of their relevance to health using relevant metrics. By health we more specifically refer to [*human health, animal health, plant health, One Health approach*].

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.4. Valuation of ecosystem services

3.4.f. Should the ecosystem service assessment evaluate the conservation, sustainable supply and / or use of the ecosystem service(s) of interest?

Answer yes, no, maybe, or don't know. Sustainable use of ecosystem services means that the services will be maintained and won't be used until depletion, but will be available for future generations. Sustainable supply of ecosystem services means that they will continue to be provided in the long term and be available for future generations. The assessment of sustainable levels of supply and use facilitates the evaluation of sustainability of use, management and conservation measures.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Since the UN Convention of Biological Diversity was opened for signature in 1992, the conservation and sustainable use of biodiversity and the benefits it provides have been the central focus of nature policy in Europe and beyond. Looking at sustainability of supply and use of ecosystem services allows for long-term planning, putting in place adaptation and mitigation strategies.

Conservation and sustainable use are critical as they link to ecosystem condition, ecosystem capacity and ecosystem service supply, which are relevant to all services.

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on sustainability and condition!

Give me an example answer

Yes, it would be relevant to assess the sustainable use of the ecosystem service especially for wood production and recreation. Important to avoid too much impact on the condition of the forest.

Terms of Reference preformatted text

The ecosystem service assessment should evaluate the sustainable [*use, supply, or both*] of these service(s), to [*if possible, precise why sustainability is important for the decision-making within which the assessment fits*].

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.4. Valuation of ecosystem services

3.4.g. Should the ecosystem service assessment include an evaluation of ecosystem condition?

Answer yes, no, maybe, or don't know. Ecosystem condition characterises the state of health of ecosystems. Ecosystem services are sensitive to a change in condition of the ecosystem that provides the service. Ecosystems in bad condition might have less capacity to provide ecosystem services.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Ecosystem condition is the state of health of ecosystems and directly influences the *supply* of ecosystem services as it determines whether or not ecosystems can effectively provide services. Indeed an ecosystem in poor condition might result in poor provision of ecosystem services such as a fewer number of services provided, or small provision of the services. While an ecosystem in good condition may lead to a higher level provision of ecosystem services.

It is important to take into account ecosystem condition together with ecosystem services to have a more complete picture and identify the potential risks in the provision of the ecosystem services and where they could occur.

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on sustainability and condition!

Give me an example answer

Yes, the assessment should include an evaluation of ecosystem condition, specifically for the forests.

Terms of Reference preformatted text

The ecosystem service assessment should include an evaluation of the condition of the ecosystem(s) providing the ecosystem service(s) of interest.

Back to the template landing page

3. GENERAL METHODOLOGICAL CHARACTERISTICS

3.5. Scenarios

3.5.a. Are future scenarios required in the ecosystem service assessment?

Answer yes, no, maybe, or don't know. Identify if there is a need to conduct the assessment while taking into account scenarios reflecting a change in policy, climate or other environmental conditions, change in governance etc. In modelling, future scenarios represent a range of potential futures that one would like to predict.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Future scenarios can be important to request if the overall purpose of the ecosystem service assessment is to inform. Broadly, scenarios help assessing ecosystem services and their associated benefits in different conditions such as policy context, climate change, land uses, land management etc. Looking at scenarios in ecosystem service assessment can also help with identifying different trade-offs between scenarios but also how trade-offs between ecosystem services and /or their benefits change over the scenarios considered.

Scenarios are a way of dealing with uncertainties about the future. They also allow to identify risk and facilitate long-term planning and adaptation.

Where can I find more questions to be more detailed in my Terms of Reference?

Visit our advanced content on uncertainty!

Give me an example answer

Yes, the assessment takes land use changes due to the implementation of the operational plan into account. This way the impact of the restoration can be measured and the actions of the plan revised if necessary.

Terms of Reference preformatted text

The ecosystem service assessment should include a scenario analysis to [*describe reason(s) why a scenario analysis should be conducted*].

Back to the template landing page

4. STAKEHOLDER ENGAGEMENT

4.a. Should the contractor conduct a stakeholder analysis?

Answer yes, no, maybe, or don't know. A stakeholder analysis aims at identifying and understanding third parties (i.e., individuals, groups, organisations/institutions) who are interested in or are expected to have an impact on the process of the ecosystem service assessment and / or its results.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Although commissioners might know which stakeholders should be involved in the ecosystem service assessment, conducting a stakeholder analysis will help with identifying other stakeholders that haven't been identified before, especially when it comes to minority and / or vulnerable social groups.

The results of a stakeholder analysis should be reviewed throughout the assessment process to take into account potential new stakeholders emerging from the results of the assessment itself.

Give me an example answer

Yes, it is interesting to do a stakeholder analysis to identify the key stakeholders.

Terms of Reference preformatted text

Prior to starting the assessment, the applicant will be asked to conduct a detailed stakeholder analysis to identify the key stakeholders that will be involved in the development of the project.

Back to the template landing page

4. STAKEHOLDER ENGAGEMENT

4.b. Is it anticipated that any stakeholder(s) will be involved at any stage of the ecosystem assessment process (*e.g.,* assessment development, practice, review, reporting)?

Answer yes, no, maybe, or don't know. Stakeholder engagement means that key parties will be consulted during the development of the assessment and the evaluation of the results and deliverables.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Stakeholders are any group, organisation or individual who are interested in and / or will or might be affected by ecosystem services and any change made to their flows. Involving key stakeholders is essential to have an inclusive and integration ecosystem service assessment. There are several other reasons including:

- For **quality insurance and knowledge integration**. Some stakeholders are data holders and their input can be very relevant for selecting the appropriate data to be used. Also, when it comes to local assessments, local stakeholders usually have good local ecological knowledge as well as a general knowledge of the area and societal contexts.
- For **legitimacy**. Involvement creates a sense of ownership of the assessment process and results. This might encourage the use of the results outside of the context of this assessment.
- For **realistic implementation**. Involving a range of relevant stakeholders means involving a range of complementary expertise and perspectives. This can help ensuring that the assessment fits within the correct governance structure and processes, but also leads to control for its feasibility notably by identifying the relevant barriers and levers.

Give me an example answer

Yes, stakeholders are expected to be involved, especially in identifying the relevant ecosystem services in National Park Mainland.

Terms of Reference preformatted text

Stakeholders are expected to be involved throughout the ecosystem service assessment. Key stakeholders who could be involved are [*include the answer to question 4.c presented on the next page*].

Back to the template landing page

4. STAKEHOLDER ENGAGEMENT

4.c. If key stakeholders have already been identified for direct involvement in the ecosystem service assessment: who are they and why have they been selected?

Identify the stakeholders who should be consulted during the development of the assessment. This can be based on an earlier stakeholder analysis or primary considerations that are important from the commissioners' perspective.

Answer in this box.

Why is this question relevant to me?

Stakeholders are any group, organisation or individual who are interested in and / or will or might be affected by ecosystem services and any change made to their flows. Stakeholders have different expertise and perspectives. It is useful to think about what the commissioners need and how stakeholders are expected to be impacted by the assessment and / or use the results of the assessment. This makes the assessment process more inclusive and ensures for quality, legitimacy and feasibility.

Give me an example answer

It is expected that the three municipalities on which the National Park Mainland lays , Nature organisations and farmers should be involved. It will enhance the integration of local knowledge and legitimacy of the results. Other scientists doing research in National Park Mainland should also be involved in order to exchange data and validate results of the assessment.

Terms of Reference preformatted text

Stakeholders are expected to be involved throughout the ecosystem service assessment. Key stakeholders who could be involved are [*list the key stakeholders who should be involved according to the commissioners' perspective and / or result from a previously conducted stakeholder analysis*].

Back to the template landing page

Methodology template

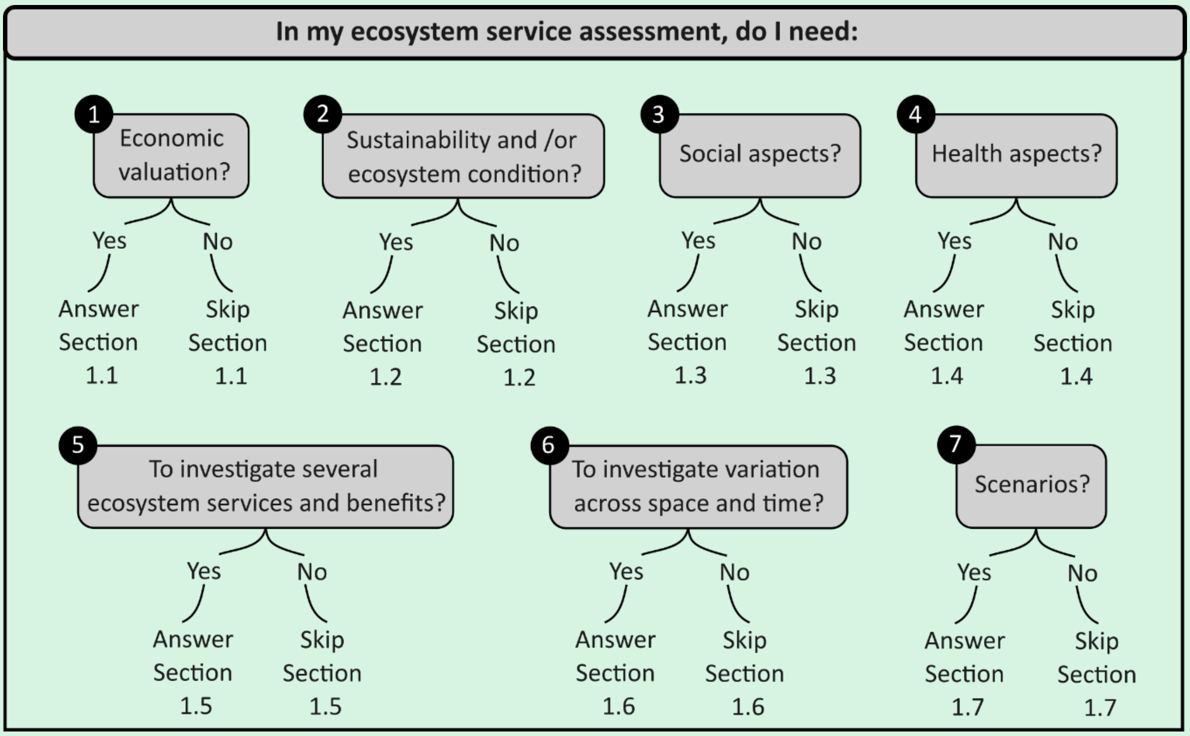
We recommend to read the 5.1 General Information prior to using the template

Below is a list of the questions included in the Methodology template. **Commissioners can use this list to select the questions that are relevant to them to write the Methodology section of a Terms of Reference.**
Each question is associated with a link to a more detailed page which will provide a box to answer the question, descriptions, examples and the preformatted Terms of Reference text that the commissioners can choose to copy-paste and modify to write the Methodology section of a Terms of Reference to commission ecosystem service assessment.

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

Describe in further details what are the technical requirements of the ecosystem service assessment. For this, commissioners can select the sections that apply to them below, and answer their corresponding questions. Including all of the sections below will result in a more exhaustive assessment than if only a few of them are included. The choice of including any of the sections below is based on the overall purpose and context of the assessment, as well as the resources available (e.g. time, budget) to the commissioners.

The commissioners can identify the sections of relevance below before moving on to the list of questions.



Back to the main landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.1. Economic valuation		**This question is relevant**
1.1.a. Which economic valuation method(s) should be applied in the ecosystem service assessment?	*Link to the detailed question*	YES / NO
1.1.b. Should the assessment include indicators of the contribution of ecosystem service(s) to economic development such as employment and growth?	*Link to the detailed question*	YES / NO
1.2. Sustainability and ecosystem condition		**This question is relevant**
1.2.a. Should the ecosystem service assessment evaluate how the condition of ecosystems influences the levels of ecosystem services supply and use?	*Link to the detailed question*	YES / NO
1.2.b. Should the ecosystem service assessment define sustainable levels for the supply and / or use of ecosystem services?	*Link to the detailed question*	YES / NO
1.2.c. What aspects of ecosystem condition are you interested in?	*Link to the detailed question*	YES / NO
1.3. Social characteristics		**This question is relevant**
1.3.a. Should the ecosystem service assessment results be disaggregated by social groups to better understand who benefits from ecosystem services?	*Link to the detailed question*	YES / NO
1.3.b. Should the ecosystem service assessment evaluate and report the attitudes, perceptions, and values of social groups towards ecosystem services?	*Link to the detailed question*	YES / NO
1.4. Health characteristics		**This question is relevant**
1.4.a. What aspects of health are you interested in?	*Link to the detailed question*	YES / NO
1.4.b. Should the results of the ecosystem service assessment evaluate and report the attitudes, perceptions, and values of social groups towards health-relevant ecosystem services?	*Link to the detailed question*	YES / NO
1.4.c. Should the ecosystem service assessment establish health links with specific ecosystem types and / or ecosystem services?	*Link to the detailed question*	YES / NO

1.5. Relationships between ecosystem services		This question is relevant
1.5.a. Should the ecosystem service assessment identify potential interrelationships between ecosystem services and / or the benefits they provide?	*Link to the detailed question*	YES / NO
1.6. Spatial and temporal characteristics		**This question is relevant**
1.6.a. Should the ecosystem service assessment evaluate how the distribution of ecosystem service(s) and / or the benefits they provide vary across space?	*Link to the detailed question*	YES / NO
1.6.b. Should the ecosystem service assessment evaluate how the distribution of ecosystem service(s) and / or the benefits they provide change over time?	*Link to the detailed question*	YES / NO
1.7. Scenarios characteristics		**This question is relevant**
1.7.a. What scenarios should be taken into account in the ecosystem service assessments?	*Link to the detailed question*	YES / NO

2. UNCERTAINTY

Describe how the uncertainties linked to the ecosystem service assessment should be assessed and reported. Uncertainty refers to the degree of confidence we can have in the results of an assessment. It includes both natural variability and gaps in knowledge or data. Quantifying uncertainties is important as these can have implications for decision-making. For this, the commissioners can answer the following questions:

		This question is relevant
2.a. Should the uncertainties inherent to the ecosystem service assessment be assessed and documented for the purpose of the decision-making?	*Link to the detailed question*	YES / NO
IF YES, ANSWER THE QUESTION BELOW		
2.b. What is the preferred reporting format to document uncertainties?	*Link to the detailed question*	YES / NO

Back to the main landing page

3. VALIDATION

Outline the expectations for validating the ecosystem service assessment. Broadly, validation means evaluating the quality of the assessment including the scope, the methodology employed and the results. For this, the commissioners can answer the following questions:

		This question is relevant
3.a. How should the methodology and results of the ecosystem service assessment be validated?	Link to the detailed question	YES / NO
IF YES, ANSWER THE QUESTION BELOW		
3.b. Should the results be compared to other published reference data in the validation process?	Link to the detailed question	YES / NO

4. STAKEHOLDER ENGAGEMENT

Identify at which steps of the ecosystem service assessment stakeholders should be informed, consulted or participate. Then, give your preference for how the stakeholders should be involved. Stakeholders are any group, organisation or individual who are interested in and / or might be affected by ecosystem services and any change made to their flows. For this, the commissioners can answer the following questions:

		This question is relevant
4.a. At which step(s) of the ecosystem service assessment should the stakeholders be informed, consulted, or participate?	Link to the detailed question	YES / NO
4.b. What methods should be used for involving stakeholders in the ecosystem service assessment?	Link to the detailed question	YES / NO

Back to the main landing page

Methodology template - detailed questions

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.1. Economic valuation

1.1.a. Which economic valuation method(s) should be applied in the ecosystem service assessment?

Select what applies. Choosing the type of primary valuation methods is voluntary, if the commissioners don't have sufficient knowledge.

Possible valuation method	Description	Valuation should be carried out this way
Value transfer methods	Value transfer is the use of results from existing primary studies to predict values or related information for other sites or policy contexts. Using value transfer methods is generally faster, less resource intensive and can be applied over larger geographic scales in comparison with primary valuation methods.	YES / NO / MAYBE / DON'T KNOW
Primary valuation methods	Primary valuation methods produce new or original information generally using primary data for the specific location of the assessment. Results directly reflect the characteristics of the ecosystem and beneficiaries.	YES / NO / MAYBE / DON'T KNOW
IF YES TO PRIMARY VALUATION METHODS, COMMISSIONERS MAY CHOOSE FROM THE METHODS BELOW:		
Revealed preference methods	Revealed preference methods observe the behaviour of beneficiaries and their use of ecosystem services to elicit values. These methods are applicable to valuing a narrow set of ecosystem services including recreation, tourism, visual amenity, and moderation of hazards (*e.g.,* flooding, air pollution).	YES / NO / MAYBE / DON'T KNOW
Stated preference methods	Stated preference methods use public surveys to ask beneficiaries to state their preferences for, generally hypothetical, changes in the provision of ecosystem services. These methods are widely applicable to value all ecosystem services but can suffer from bias due to survey design and the hypothetical nature of recorded preferences.	YES / NO / MAYBE / DON'T KNOW
Market prices	Prices of some ecosystem services can be obtained from markets or surveys of businesses and households. This approach is only applicable to ecosystem services that are traded in markets (*e.g.,* extracted resources, carbon credits)	YES / NO / MAYBE / DON'T KNOW
Replacement cost	The replacement cost estimates the value of an ecosystem services as the cost of replacing it with a human-made service. It is only applicable to ecosystem services that have human-made alternatives (e.g., coastal protection by seawalls, water supply by reservoirs).	YES / NO / MAYBE / DON'T KNOW
Damage cost avoided	The damage cost avoided method estimates the value of ecosystem services that provide some form of protection (*e.g.,* mitigation of floods, filtration of pollutants, climate regulation) as the cost of damage that is avoided.	YES / NO / MAYBE / DON'T KNOW

Social cost of carbon	The social cost of carbon (SCC) is a special case of the damage cost avoided method used to value global climate regulation. The SCC is the monetary value of damages caused by emitting one tonne of CO2 in a given year. The SCC therefore also represents the value of damages avoided by reducing the emission or sequestering one tonne of CO2.	YES / NO / MAYBE / DON'T KNOW
Net factor income	The net factor income method estimates the value of ecosystem services that are inputs into the production of marketed goods (e.g., filtration of drinking water, nursery for fisheries) as revenue from sales of the good minus the cost of other inputs.	YES / NO / MAYBE / DON'T KNOW
Production function	The production function method estimates the value of ecosystem services that are inputs into the production of marketed goods (e.g., filtration of drinking water, nursery for fisheries) by estimating a statistical function that models the value of different inputs into production.	YES / NO / MAYBE / DON'T KNOW
Hedonic pricing	Hedonic pricing is a revealed preference method that estimates the influence of ecosystem services on the prices of marketed goods (usually residential property). It is applicable for valuing recreational use, visual amenity and filtration of air pollution by urban green.	YES / NO / MAYBE / DON'T KNOW
Travel cost method	The travel cost method estimates the value of nature based recreation using data on travel costs and visit rates.	YES / NO / MAYBE / DON'T KNOW
Contingent valuation	Contingent valuation involves asking beneficiaries to state their willingness to pay for ecosystem services through public surveys. This method can be applied to value all ecosystem services.	YES / NO / MAYBE / DON'T KNOW
Choice modelling	Choice modelling (or discrete choice experiments) involves asking beneficiaries to state their willingness to pay for ecosystem services by making choices between alternative levels of provision and payment. This method can be applied to value all ecosystem services.	YES / NO / MAYBE / DON'T KNOW
Other (specify)		YES / NO / MAYBE / DON'T KNOW

Why is this question relevant to me?

Economic valuation methods include a wide range of approaches for estimating the contribution of ecosystem services to human well-being. This question is relevant if the commissioners wish to specify which economic valuation methods should be applied in the ecosystem service assessment. The choice of which valuation method(s) should be applied is dependent on the ecosystem services to be valued (some methods are directly relevant to specific services), data and resource availability, and the required level of certainty and site-specificity of the results.

A first important choice is between "primary valuation methods" and "value transfer methods". The former set of methods produce new or original information generally using primary data for the study area whereas the latter set of methods use existing information from valuation studies conducted elsewhere and make adjustments to reflect the characteristics of the study site.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on economic valuation!

Give me an example answer

Considering our knowledge of economic valuation and the foreseen budgets, we would use value transfer methods, adjusted to the characteristics of the study site.

Terms of Reference preformatted text

[*if value transfer methods are chosen*]

The study should design and apply value transfer methods to estimate the economic value of each identified ecosystem service. Transferred values should be adjusted to reflect the characteristics of the study site in terms of ecosystem extent, condition, population of beneficiaries and income level. This could involve the use of available databases of ecosystem service values and / or the use of published value transfer functions.

[*if primary valuation methods are chosen*]

The ecosystem service assessment should design and apply [*relevant primary valuation methods, OR add the primary valuation methods chosen if any*] to estimate the economic value of the ecosystem service(s) of interest, including the collection of required data through surveys of beneficiaries in the study area (if any).

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.1. Economic valuation

1.1.b. Should the assessment include indicators of the contribution of ecosystem service(s) to economic development such as employment and growth?

Answer yes, no, maybe, or don't know. If yes and if commissioners have the knowledge, they can specify the type of economic indicator of interest.

The economic importance of ecosystem services can be quantified through indicators such as the contribution to economic development, sectors of the economy, or employment. These indicators can be useful for communicating the importance of ecosystem services to decision makers and the general public.

Answer in this box (yes, no, maybe, or don't know). If yes, specify.

Why is this question relevant to me?

This question is relevant if commissioners want the ecosystem service assessment to quantify indicators of economic importance. Potentially relevant indicators include the contribution of ecosystem services to economic development, specific sectors of the economy, or local employment. These indicators can be useful for communicating the importance of ecosystem services to decision makers and the general public.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on economic valuation!

Give me an example answer

Yes. Measures of the contribution of the ecosystem services to the economic importance should be complied, particularly indicators for recreation sector and local employment.

Terms of Reference preformatted text

The ecosystem service assessment should measure the contribution of the ecosystem service(s) of interest to the economic development of the area of interest. [*Particular economic indicators of interest are…, OR In their proposal, the contractor should clearly describe and justify the economic indicators that will be used*].

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.2. Sustainability and ecosystem condition

1.2.a. Should the ecosystem service assessment evaluate how the condition of ecosystems influences the levels of ecosystem services supply and use?

Answer yes, no, maybe, or don't know. Ecosystem condition relates to the quality of an ecosystem. The levels of ecosystem service(s) supply and use may vary depending on changes in condition of the ecosystem that provides the service. Ecosystem service(s) supply is the provision of service(s) by the ecosystem(s). Ecosystem service(s) use is the utilisation of service(s) by those who benefit from them.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Ecosystem condition assessments are based on the evaluation of the levels of ecosystem properties. The properties of the ecosystem will, in turn, determine the levels of ecosystem services supply and use. Degraded ecosystems may have lost their capacity to supply ecosystem services. Maintaining ecosystems in good condition is central to sustainability. This question is relevant if we wish to assess, for example, the consequences of protecting ecosystems from current levels of use, or of ecosystem restoration actions in terms of ecosystem services supply and use.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on sustainability and condition!

Give me an example answer

Maybe. As we ask for a forest ecosystem condition it would be nice to link the forest condition to the supply of ecosystem services, depending on the variables used in the biophysical assessment.

Terms of Reference preformatted text

The ecosystem service assessment should include analyses to highlight and understand the influence of ecosystem condition on the [*supply, use*] of ecosystem service(s) of interest. In their proposal, the contractor will clearly describe and justify the methodology that will be used.

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.2. Sustainability and ecosystem condition

1.2.b. Should the ecosystem service assessment evaluate and define sustainable levels for the use and / or supply of ecosystem services?

Answer yes, no, maybe, or don't know. Sustainable use of ecosystem services means that the ecosystem services won't be used until depletion and will still be available for future generations. Sustainable supply of ecosystem services means that they will continue to be provided long-term and be available for future generations. Defining sustainable levels refers to the evaluation and identification of maximum use that is compatible with the supply of ecosystem services. If these levels are respected, the ecosystem condition and ability to deliver services for future generations is maintained.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Protecting and preserving the biodiversity that underpins ecosystem services helps maintain ecosystems resilience and ensures long-term service provision. Evaluating and defining the need for protection of ecosystems and their restoration along with levels of use ensure that ecosystem services are not overused, helping to maintain the ability of ecosystems to provide ecosystem services for future generations. It supports informed decision-making for long-term ecosystem health and resilience.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on sustainability and condition!

Give me an example answer

Yes, especially the sustainable use of wood needs to be analysed.

Terms of Reference preformatted text

The ecosystem service assessment should define the sustainable levels for [*supply, use*] of the ecosystem service(s) of interest. In their proposal, the contractor will clearly describe and justify the methodology that will be used to identify and define these levels.

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.2. Sustainability and ecosystem condition

1.2.c. What aspects of ecosystem condition are you interested in?

Ecosystem condition encompasses a lot of different dimensions. Select those of interest in this assessment.

Dimension of ecosystem condition	Description	This dimension should be measured in the assessment
Index of general ecosystem condition	This kind of index is an overarching measure of ecosystem condition. It is usually a combination of all or some of the indicators mentioned below. It is useful to give an overview of the condition of an ecosystem. It can also help in broad scale assessments of the linkages between levels of pressures and use, and trade-off analysis between ecosystem condition and ecosystem services.	YES / NO / MAYBE / DON'T KNOW
Biotic indicators - Species	Species condition indicators show how well species are doing in an ecosystem such as the size of the population, the reproductive success or their spatial distribution within the ecosystem. It can be very important to measure them to identify whether the level of use of ecosystem service exceeds the supply. For instance, species condition indicators based on population recruitment will help determine whether harvest levels in fisheries and game are sustainable.	YES / NO / MAYBE / DON'T KNOW
Biotic indicators - structure	Structural condition indicators show how well the physical structure of an ecosystem (*i.e., whether a forest consists of tree, shrub and herb layers, and dead wood*) is doing. For example, the amount of dead wood can inform on the supply of ecosystem services such as habitat provision for native biodiversity and vegetation cover on soil retention and erosion control.	YES / NO / MAYBE / DON'T KNOW
Biotic indicators - function	Functional condition indicators can inform about the level of specific ecological functions and processes of an ecosystem. These indicators are based on identifying and grouping species with similar ecological functional characteristics. For example, pollinators, organic matter decomposers, and herbivores. Diversity of functional groups is related to ecosystems' multifunctionality, and the potential supply of multiple ecosystem services. Diversity within functional groups can ensure ecosystem resilience, which is key in facilitating sustainable supply of ecosystem services.	YES / NO / MAYBE / DON'T KNOW
Abiotic - Physical indicators	Abiotic physical condition indicators show levels of non-living physical components of an ecosystem such as the pH of the soil, or the	YES / NO / MAYBE / DON'T KNOW

	salinity and temperature of water. If affected, they can have an impact on the supply of ecosystem services. Abiotic physical indicators include for instance, the levels of the water table of wetlands and the water holding capacity of the soil. If these levels change, this will result in an impact on ecosystem services provided by wetlands including changes in carbon storage and resilience to drought spells of croplands, respectively.	
Abiotic - Chemical indicators	Abiotic chemical condition indicators show the chemical properties in water, soil and air of an ecosystem. Examples are levels of nitrates in soil and water through the addition of fertilizers, or particulate matter in the air through pollution. Changes in the chemical properties of an ecosystem impact the supply of their services such as habitat provision for biodiversity. These indicators can help with identifying areas with chemical imbalances to be addressed, or target areas for depollution..	YES / NO / MAYBE / DON'T KNOW
Landscape-level characteristics	Landscape condition indicators help assess how an ecosystem is physically organised in space. They help with identifying whether ecosystems are fragmented in space, or support a good connectivity allowing species to move between different patches. This is notably critical to ensure the supply of ecosystem services such as habitat provision for biodiversity. These indicators are particularly helpful to guide landscape level land-use planning, conservation and restoration.	YES / NO / MAYBE / DON'T KNOW

Why is this question relevant to me?

Ecosystem condition indicators represent properties of an ecosystem. They allow to evaluate which aspects of an ecosystem are in good or bad condition. As there are different families of condition indicators measuring different aspects of an ecosystem condition, it is important to measure several types of condition properties to have a holistic overview of the condition of an ecosystem. On the other hand, measuring specific ecosystem properties (e.g. population recruitment, occurrence of functional groups, chemical properties), can help establish cause-effect relationships between ecosystem properties and levels of ecosystem service supply, thus enabling quantitative assessments of ecosystem services change.

To summarise several indicators, they can be assembled in so-called indices. These indices are composite measures representing the overall state or condition of an ecosystem. Indices are particularly useful when the purpose is to assess how the general ecosystem condition state relates to the levels of supply of ecosystem service(s).

Indicators and indices can be compared to values in a reference state of condition. Such comparisons help provide a general overview of how the properties of an ecosystem have evolved compared to a known reference state, thus indicating an increase or a decrease. These comparisons are useful to, for example, assess the general need for ecosystem restoration or to measure the effect of restoration actions.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on sustainability and condition!

Give me an example answer

The evaluation of ecosystem condition in this assessment should encompass the following aspects: biotic and abiotic indicators, fragmentation and combine this in a general index.

Terms of Reference preformatted text

The evaluation of ecosystem condition in this assessment should encompass the following aspects [*choose what is relevant from the list below*]:

- An index of general ecosystem condition covering at least [*biotic condition indicator(s) for species composition, biotic condition indicator(s) for the structure of an ecosystem, biotic condition indicator(s) for the functions of an ecosystem, abiotic condition indicator(s) for the physical state of an ecosystem, abiotic condition indicator(s) for the chemical state of an ecosystem, condition indicator(s) for the landscape-level characteristics of an ecosystem*].
- Biotic condition indicators for [*species composition, ecosystem structure, ecosystem functions]*
- Abiotic condition indicator(s) for the *[physical, chemical characteristics of the ecosystem*]
- Landscape characteristics indicator(s)

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.3. Social characteristics

1.3.a. Should the ecosystem service assessment results be disaggregated by social groups to better understand who benefits from ecosystem services?

Answer yes, no, maybe, or don't know. Understanding who benefits from the ecosystem services guarantees that inequities are identified and addressed. This approach highlights who benefits, who is excluded, and how interventions can improve well-being and equity. Examples of social groups are: gender, age, race, ethnicity, (dis)ability, income status, etc.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Identifying social groups most reliant on ecosystem services highlights the populations at greatest risk from any reduction in access or supply. Understanding access barriers provides actionable insights for addressing inequities and ensuring that specific groups are not excluded from the benefits of ecosystem services.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on social benefits!

Give me an example answer

No. The focus lays on the supply of ecosystem services rather than demand.

Terms of Reference preformatted text

The ecosystem service assessment should measure the distribution of the ecosystem service(s) of interest by social groups or variables (e.g. income, age, gender). Dependency(ies) on and / or access(es) to ecosystem services should be clearly reported using measurable indicators (e.g., demand, proximity, frequency use, and availability) to identify which groups face the greatest barriers or are most dependent on specific ecosystem services. In their proposal, the contractor will clearly describe and justify the methodology (including indicators and metrics) that will be used.

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.3. Social characteristics

1.3.b. Should the ecosystem service assessment evaluate and report the attitudes, perceptions, and values of social groups towards ecosystem services?

Answer yes, no, maybe, or don't know. This allows to investigate whether the ecosystem service assessment will document how different social groups perceive, value, and prioritize ecosystem services.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Attitudes, perceptions, and values provide insights into how ecosystem services are viewed and prioritized. This helps align the assessment with the needs/ expectations of (relevant) stakeholders/actors. This is of particular relevance if the assessment aims at delivering recommendations.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on social benefits!

Give me an example answer

Yes, stakeholders will be involved in the selection of relevant ecosystem services.

Terms of Reference preformatted text

The ecosystem service assessment should report the attitudes, perceptions, and values of social groups toward the ecosystem service(s) of interest. The results should lead to deliver actionable insights based on stakeholder preference. In their proposal, the contractor will clearly describe and justify the methodology that will be used.

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.4. Health characteristics

1.4.a. What aspects of health are you interested in?

Select what applies.

Aspects of ecosystem services that can be measured	Description	This aspect should be measured
Health benefits	Health benefits include goods or services which promote good health or offer avenues for positive health interventions. Considering health benefits (*e.g.,* supplied by salutogenic environments, or provided by medicinal resources) in an ecosystem service assessment can identify important resources for maintaining or enhancing health. For example, access to high quality green areas in cities can help to reduce stress and encourage physical activity among urban residents.	YES / NO / MAYBE / DON'T KNOW
Health risks	Health risks refers to potential threats to good health, including drivers of disease emergence. Considering actual or potential health risks (*e.g.,* of exposure to pathogens or natural hazards) in an ecosystem service assessment can greatly enhance health promotion efforts and related prevention and preparedness efforts. For example, changes in ecosystem structure and function can alter the ecology of infectious diseases, leading to increased risk of disease emergence in wild or domesticated species, and humans. Another example is how loss of coastal ecosystem condition can increase exposure of coastal communities to risk of coastal flooding, storm surges and tsunamis, with significant, social and economic implications.	YES / NO / MAYBE / DON'T KNOW
Health outcomes	Health outcomes refers to the status of health as a result of an intervention or exposure to one or more conditions. Considering health outcomes (*e.g.,* the incidence or prevalence of positive or negative health status at varying population scales, and / or across time) in an ecosystem service assessment can help to quantify the social, cultural and economic aspects of ecosystem condition and ecosystem change. While health risks and benefits refer generally to the ways in which ecosystem types and /or services may impact or improve health, health outcomes refer more specifically to the measurable health effects associated with access to (or changes to) ecosystem services. For example, an assessment of health outcomes arising from an increase in high quality green and blue urban environments might quantify reductions in mental health visits to primary care services in a specific population across multiple years. Another example would be a determination of the	YES / NO / MAYBE / DON'T KNOW

	rise or fall in incidence of water-borne disease in a specific community within a specific period after changes to aquatic ecosystems used for potable water supply.	

Why is this question relevant to me?

Assessments considering health benefits, risks, and / or outcomes can identify the likelihood and consequences of impacts associated with specific development or management strategies. However, the criteria by which ecosystem-health relationships are to be assessed will inform the selection of assessment methodology, stakeholder engagement and reporting, and the utility of the assessment outputs.

For example, a generalised assessment of health benefits or risks might focus on qualitative methods such as community self-reporting as well as a general understanding of the relationships between health and specific ecosystem services, and the opinions of health care providers and civil society organisations, whilst an assessment of health outcomes may require more detailed use of epidemiological methods and engagement with medical scientists or health economists. General assessments of benefits or risks may be most useful in broad communications of the linkages between ecosystems and health, for public engagement or wider mainstreaming efforts, and to provide a general health context for decision makers. Detailed assessments of health outcomes can help quantify costs and risks in social, cultural, and / or economic terms, inform health strategies and action plans and design of targeted interventions, or support the development of a more robust evidence base for the link between ecosystems and well-being and the implications of environmental change.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on health benefits!

Give me an example answer

Yes, the three aspects of health should be considered if possible and if data are available. Health benefits of interest are more specifically the potential for reduction of stress and improvement of mental health associated with access to the natural area of National Park Mainland.

Terms of Reference preformatted text

The ecosystem service assessment should consider the health dimension. In particular it should include an evaluation of health [*benefits, risks, outcomes*].

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.4. Health characteristics

1.4.b. Should the ecosystem service assessment evaluate and report the attitudes, perceptions, and values of social groups towards health-relevant ecosystem services?

Answer yes, no, maybe, or don't know. Investigate whether the ecosystem service assessment will document how different social groups perceive, value, and prioritize ecosystem services. These perceptions may include cultural significance, recreational use, or economic reliance.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

The relationships between ecosystems and health can be highly dependent upon prevailing social and cultural contexts, traditions, worldviews or experiences. Understanding diverse perceptions ensures that the assessment results better account for community values and cultural priorities, and can add important detail to the assessment of health risks, benefits or outcomes. it is also key to ensuring that assessments of health-relevant ecosystem services address issues of health equity, and can also help avoid or resolve conflicts by revealing broader contexts and balancing competing interests.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on health benefits!

Give me an example answer

Yes, the consideration of the health benefits of National Park Mainland should consider the ability of different age groups to access the areas as well as their specific preferences in terms of the location, routes of access to, and overall design of green areas.

Terms of Reference preformatted text

The ecosystem service assessment should capture social groups' attitudes, perceptions, and values toward specific ecosystem services of relevance to health, focusing on their social, cultural, and economic importance.

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.4. Health characteristics

1.4.c. Should the ecosystem service assessment establish health links with specific ecosystem types and / or ecosystem services?

Answer yes, no, maybe, or don't know. This determines whether the ecosystem service assessment will consider general relationships between environmental compartments (e.g. air, water, soil, urban environments, rural landscapes etc.) and health, or aim to provide a more detailed understanding of the impact of defined ecosystems types and / or their services on specific health issues.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

This seeks to clarify the extent to which an assessment should explore the relationships between ecosystems, their services and health, which in turn will inform stakeholder engagement and the selection of assessment methods. It asks commissioners to decide whether an assessment should identify general, non-specific connections between the natural environment and health (*e.g.,* "managed forests provide recreational opportunities which can support good mental and physical well-being") or aim to explore more detailed cause-effect relationships between specific ecosystems and health (*e.g.,* "mixed native woodland ecosystems are preferred by a majority of outdoor enthusiasts for their diversity of flora and fauna, which visitors describe as being important for relaxation and inspiration and therefore more attractive for physical activity such as hiking and cycling").

The decision will usually be informed by the policy, social and cultural context of the assessment, and depend upon the intended audiences for the results. Detailed assessments may be important for public health planners to identify ecological drivers of health status, to inform health policies and markets, or to strengthen the evidence base for nature-based solutions. However, a more general assessment may be preferred if the aim is to provide signposts for future research and policy, or to help mainstream ecosystem service approaches into the health sector at a higher level.
The choice will influence the methodological approach, notably the type and scope of data and the appropriate metrics and indicators to use, and guide engagement with stakeholders and other experts.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on health benefits!

Give me an example answer

Yes, the assessment will seek to provide evidence for the specific ecosystems or ecosystem services which may influence the health of the local community.

Terms of Reference preformatted text

Linkages between [*health risks, benefits, outcomes*] and the ecosystem services and / or types focused on in the ecosystem service assessment should be identified. The strength, direction and / or sustainability of these linkages should also be clearly highlighted and described.

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.5 Relationships between ecosystem services

1.5.a Should the ecosystem service assessment identify potential interrelationships between ecosystem services and / or the benefits they provide?

Answer yes, no, maybe, or don't know. Interrelationships between ecosystem services, between benefits, or between ecosystem services and their benefits means that the services and / or benefits are related and affect each other. There are a number of ways ecosystems services and / or benefits can be related such as interdependencies, synergies, trade-offs…

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

There are a number of ways by which ecosystem services and /or their benefits can be linked to each other. Some examples are:

- Interdependency: this means that ecosystem services and / or their benefits rely on other ecosystem services and / or benefits. For example, the provision of pollinator-dependent food depends on the pollination service, which in turn depends on habitat provision services (*i.e.,* habitats providing nesting and flower resources). An example of interdependency between a benefit and an ecosystem service would be the mental and physical health benefit supplied by green spaces which depend on the recreation service these areas can provide.
- Trade-offs: this means that improving one service and / or a benefit might reduce one or several service(s) and / or benefit(s), because they are responding to the same driver of change, or because they are causal related. For example, draining wetlands for agriculture increases the provision of food but decreases flood mitigation, water filtration and retention, and carbon storage services. An example of trade-offs between ecosystem services and benefits would be the increase of recreational access to a natural area which might impact its cultural or spiritual significance to local communities, which in turn can have negative implications for social and mental health.
- Synergies: this means that enhancing one ecosystem service will enhance (an)other ecosystem service(s). For example, coral reefs provide habitat maintenance for native biodiversity and nursery services to fisheries at the same time that their conservation provides coastal protection. Likewise for benefits, synergies mean that receiving one benefit will strengthen another benefit. For example, urban green spaces supply urban cooling which provides health benefits by reducing heat-related illnesses, this in turn also provides a social benefit by improving social well-being.

These relationships between ecosystem services and between the benefits they provide are crucial to take into account for decision-making. They are necessary to understand when assessing sustainability, as well as for identifying and analysing situations where maximizing certain benefits may inadvertently diminish others.

Recognising trade-offs between services helps understanding, in a holistic way, the possible impacts of a certain intervention aiming at one service on other services. Such evaluations also help balance competing priorities, ensuring equitable access to benefits.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on sustainability and condition, as well a social benefits and health benefits!

Give me an example answer

Yes. We would like to have an insight in the synergies and trade-offs between ecosystem services.

Terms of Reference preformatted text

The ecosystem service assessment should identify and quantify the strength of potential interrelationships between the [*ecosystem service(s) of interest, benefits provided by ecosystem services, ecosystem services and the benefits they provide*]. In their proposal, the contractor will clearly describe and justify the methodology that will be used to identify and quantify these trade-offs.

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.6. Spatial and temporal characteristics

1.6.a. Should the ecosystem service assessment evaluate how the distribution of ecosystem service(s) and the benefits they provide vary across the space?

Answer yes, no, maybe, or don't know. This helps determine whether the assessment should quantify and track how the ecosystem service supply, use, demand but also the benefits derived from them vary across the geographic areas and spatial scales (e.g., continental, national, regional, local) considered in the assessment. For example, this could allow to identify differences in supply and / or benefits between different countries, regions, municipalities, or other relevant areas.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Ecosystem services are the ecological process by which benefits are provided. For example water filtration, urban cooling, and pollination of crops are ecosystem services, while the associated benefits are the availability of clean drinking water, the reduction of heat-related illnesses, and food production.

Ecosystem services are not evenly distributed across geographic areas and spatial scales. This means that their supply, use, demand but also associated benefits vary across both geographic areas and spatial scales. Studying the spatial variation of the ecosystem services supply support decision-making by identifying where the services are available to beneficiaries and where they are not. Overlapping this with the spatial distribution of use and demand also allows to highlight whether the demand for ecosystem services meet the actual need for them. This also allows to reveal trends in supply and access, supporting strategic decisions on sustainable supply and use of the services. From a social benefits perspective, this is particularly relevant as it helps identifying underserved locations and prioritize interventions where they have the greatest impact.

Regarding health benefits, accounting for the spatial variation is crucial as these benefits can occur outside of the spatial scale and geographical area in which the ecosystem types and their service are. For example, the restoration of wetlands enhances the capacity to retain water and mitigate floodings, which in turn reduces incidence of waterborne diseases and disaster fatalities in downstream communities due to avoided floodings.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on spatial and temporal characteristics, as well a social benefits and health benefits!

Give me an example answer

Yes. The assessment should look for the supply of ecosystem services only to the geographical scale of the National Park Mainland. For the benefits such as recreation and climate mitigation, it will go beyond this scale as the area attracts visitors from outside the geographical scale of the National Park Mainland. Also, the effects of climate mitigation are not necessary visible on the local scale.

Terms of Reference preformatted text

The assessment should evaluate the spatial variations in the levels of [*supply, use, demand*] of the ecosystem service(s) of interest [*and their associated benefits*] over the spatial extent and scales considered in this assessment. Results should highlight the trends and identify key drivers of change. [*if relevant, add the following sentence*] These variations will have to be specifically put in relation with sustainable [*supply, use*]. In their proposal, the contractor will clearly describe and justify the methodology that will be used.

[*if relevant, add the following sentence*] The assessment should also provide recommendations for maintaining or enhancing the [*supply, use*] of the ecosystem service(s) of interest [*and their associated benefits*] without negatively affecting the ecosystem condition.

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.6. Spatial and temporal characteristics

1.6.b. Should the ecosystem service assessment evaluate how the distribution of ecosystem service(s) and the benefits they provide change over time?

Answer yes, no, maybe, or don't know. This allows for quantifying and tracking how the ecosystem service(s) supply, use, demand but also the benefits derived from them vary over time due to management actions, environmental factors, or policy changes.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

The supply, use and demand of ecosystem services are not constant over time. Depending on the selected ecosystem services, it may be reasonable to consider temporal fluctuations in the assessment.

For example, the pollination service in European countries is only supplied over late spring to the beginning of fall. This means that pollinator-dependent crops for food production will be strongly dependent on seasonality. In some regions, due to a combination of climatic factors, there can be a mismatch between the flowering period and the time interval when pollinators are active. Assessing the activity patterns of pollinator species along the season will determine temporal variation of pollination services for different crops. Studying this temporal variation of the ecosystem services supports decision-making by identifying when the services are available to beneficiaries and when they are not. This also allows to reveal trends in supply and access, supporting strategic decisions on sustainable supply and use of the services.

From a social benefits perspective, it is particularly relevant to consider changes over time as it ensures that strategies are designed to sustain or enhance social benefits in the long term. Likewise, monitoring changes in health benefits over time helps decision-makers understand the long-term impacts of ecosystem service management on societal well-being and adjust strategies accordingly.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on spatial and temporal characteristics, as well as social benefits and health benefits!

Give me an example answer

Yes, the intention of the assessment is to see how the actions taken in the National Park Mainland help in reaching the goals of the Masterplan. The assessment should evaluate the temporal variations in the level of supply of the ecosystem services and related benefits over the time period and temporal scale considered in the assessment.

Terms of Reference preformatted text

The assessment should evaluate the temporal variations in the levels of [*supply, use, demand*] of the ecosystem service(s) of interest [*and their associated benefits*] over the time period and temporal scale(s) considered in this assessment. Results should highlight trends and identify key drivers of change. In their proposal, the contractor will clearly describe and justify the methodology that will be used.

[*if relevant, add the following sentence*] The assessment should also provide recommendations for maintaining or enhancing the [*supply, use*] of the ecosystem service(s) of interest [*and their associated benefits*] without negatively affecting the ecosystem condition. [*if relevant, add the following sentence*] These variations will have to be specifically put in relation with sustainable [*supply, use*].

Back to the template landing page

1. METHODOLOGICAL CHARACTERISTICS OF THE OUTPUTS

1.7. Scenarios

1.7.a. What scenarios should be taken into account in the ecosystem service assessments?

Identify the previously chosen metrics for which a scenario analysis is required. List and describe the scenarios that should be taken into account. Even a broad description of goals for which the scenarios are required is sufficient. In modelling, future scenarios represent a range of potential futures that one would like to predict.

Answer in this box.

Why is this question relevant to me?

Future scenarios can be important to request if the overall purpose of the ecosystem service assessment is to inform, guide or to request for recommendations. They are a way of dealing with uncertainty about the future. Scenarios allow to identify risk and facilitate long-term planning and adaptation.

There are different types of scenarios such as:

- land use and land cover change including restoration, deforestation management plans, etc.;
- climate change including different level of global warming, changes in rainfall and drought patterns, etc.
- changes in policy and governance including environmental policies and incentives, etc.;
- disaster risk reduction and resilience including adaptative planning to extreme events, development of green infrastructure and nature-based solutions, etc;
- etc.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on uncertainty!

Give me an example answer

The assessment needs to consider land use and land cover changes and development of nature based solutions.

Terms of Reference preformatted text

The ecosystem service assessment should include a scenario analysis to [*reason(s) why a scenario analysis should be conducted*]. Scenarios that should be considered are:

- [*brief description of scenario 1*]
- [*brief description of scenario 2*]
- […]

In their proposal, the contractor will clearly describe and justify the scenarios and methodology that will be used.

Back to the template landing page

2. UNCERTAINTY

2.a. Should the uncertainties inherent to the ecosystem service assessment be assessed and documented for the purpose of the decision-making?

Answer yes, no, maybe, or don't know. Uncertainty is present in all ecosystem service assessments. It is essential that both the commissioners and potential contractors are conscious of the sources of uncertainty that matter for a particular purpose. We recommend that commissioners ask for the assessment and documentation of uncertainties pertaining to any results of their requested assessment.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Various uncertainties are inherent in the assessment of ecosystem services and their translation to decision-makers. This means that any results and decisions taken from them will be associated with a degree of uncertainty. Uncertainties have different sources, they can come from the data sources used, the choice of the model and their parameters, as well as the interpretation that commissioners will have of the results. Broadly, Uncertainties can be classified in three families: decision uncertainties, model uncertainties, scenario uncertainties. For more information refer to section 5.1.5.4 Uncertainty.

Informed decision-making needs to take into account these uncertainties as they pertain to the robustness, relevance, and reliability of the results upon which they rely.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on uncertainty!

Give me an example answer

Yes.

Terms of Reference preformatted text

The ecosystem service assessment should evaluate and document uncertainty arising in the assessment. The contractor should clearly discuss the reliability of the methods and results in the context of their uncertainty and what it means for the decision-making that would arise from any of the results of this assessment. When reporting uncertainties, the contractor should clearly explain and justify the choice of methods used.

Back to the template landing page

2. UNCERTAINTY

2.b. What is the preferred reporting format to document uncertainties?

Select what applies.

Possible formats	Why should I choose this format?	I would like this format
Uncertainty section/report	Ask the potential contractors to document uncertainties in a separate section of the main report or as a separate short deliverable. A structured report can use clear language and summary tables to highlight uncertainties across results, data sources, and models.	YES / NO / MAYBE / DON'T KNOW
Visual summary	Contractors should report uncertainties using visual tools like confidence maps, interval bars in plots, or a traffic light system (green-orange-red) to show result reliability. These visual cues make uncertainties easier to understand, especially for those with less experience.	YES / NO / MAYBE / DON'T KNOW
Decision-relevant summary	These summaries are designed for decision-making and can take different forms based on commissioners' needs. Options include **executive highlights** for key uncertainties, **risk profiles** outlining risks in management choices, and an **assumption register** listing critical assumptions and their impacts.	YES / NO / MAYBE / DON'T KNOW
Scenario-based documentation	This documentation is relevant only if a scenario analysis is requested. Commissioners can ask for comparative tables, best- and worst-case outcomes, or qualitative scenario narratives to explore uncertainties and alternative results.	YES / NO / MAYBE / DON'T KNOW
Participatory documentation	This is a basic, qualitative approach to documenting uncertainties, relying on expert and stakeholder judgments. Engaging stakeholders helps capture their perceptions of uncertainties and their potential impact on results.	YES / NO / MAYBE / DON'T KNOW
Technical appendices	Commissioners can request advanced technical documentation as appendices, detailing data, models, and uncertainty analysis. This should include model structure, parameters, assumptions, and sensitivity analysis to show how input changes affect results.	YES / NO / MAYBE / DON'T KNOW
Other (specify)		YES / NO / MAYBE / DON'T KNOW

Why is this question relevant to me?

Documenting uncertainty can be costly, but neglecting it may lead to even greater costs in the long term, potentially resulting in unintended consequences and unfavourable outcomes. Sources of uncertainties are numerous, potential contractors need to be aware of the expectations of the commissioners and what the results will be used for to choose the best assessment and reporting of uncertainties.

Being aware of the minimum documentation requirements to support a decision, and having a conscious approach to whether there are uncertainty thresholds that may stop a decision, is part of rational decision-making.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on uncertainty!

Give me an example answer

Uncertainty can be reported in the format of a Participatory documentation based on expert judgement. We aim for a qualitative assessment of uncertainties. Assumptions taken in the modelling need to be clearly reported.

Terms of Reference preformatted text

The ecosystem service assessment should evaluate and document uncertainty arising in the assessment. The contractor should clearly discuss the reliability of the methods and results in the context of their uncertainty and what it means for the decision-making that would arise from any of the results of this assessment. The preferred format to report such uncertainties are: [*list the formats of relevance*].

When reporting uncertainties, the contractor should clearly explain and justify the choice of methods used.

Back to the template landing page

3. VALIDATION

3.a. How should the methodology and results of the ecosystem service assessment be validated?

Select what applies. Validating the methodology means ensuring that the metrics, data, choice of scenarios and analyses, and stakeholder involvement plan chosen meet the goals of the commissioners. Validating the results of the assessment means verifying that the findings are accurate, reliable, and answer the goals of the commissioners.

Validation process	Description	Validation should be carried out this way
Internal validation of the methodology and / or results	The validation process will not involve third parties such as external stakeholders. Examples of internal validation of the methodology can be planning a pilot to test the methodology before scaling it up. It can also be peer-reviews within the contractor's team of the assumptions, models or code that are planned to be used. More simply, it can be indicating that there will be ground-truthing. Examples of internal validation of the results can be peer-review within the contractor's team, or comparison with similar ecosystem service assessment conducted previously. It can also include the use of published reference data to compare how the results differ from publicly acknowledged references or benchmarks.	YES / NO / MAYBE / DON'T KNOW
External validation of the methodology and / or results	The validation process will involve external third parties such as stakeholders. Stakeholders can be involved in numerous ways such as advisory boards, public consultations or workshops. **Section 4. Stakeholder Engagement** expands more on this.	YES / NO / MAYBE / DON'T KNOW

Why is this question relevant to me?

Validating the methodology helps avoiding biases, ensures transparency, and confirms that appropriate methods are used to meet the commissioners' goals.

Validating results ensures that the conclusions drawn are based on accurate, representative data. This step is essential to avoid misleading commissioners and other potential stakeholders with incorrect information that could lead to improper decision-making.

Validation improves credibility, transparency and accountability in an ecosystem service assessment. It ensures that the assessment is accurate, trustworthy, and applicable to real-world decision-making. By validating an ecosystem service assessment at different steps, we also ensure comparability with previous work done, and can identify important caveats and limitations that commissioners and potential other stakeholders should be aware of when using the results of the assessment.

Involving third parties in the validation process can be more costly as it implies using more resources such as time and organisation of meetings and events. It is important for the contractor to know how the commissioners would prefer the validation to be carried out to prepare the workplan accordingly and select relevant stakeholders if required.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on uncertainty!

Give me an example answer

Validation of the results will be done internally and externally. Internally by comparing the results with available literature and externally through a workshop with different stakeholders.

Terms of Reference preformatted text

The ecosystem service assessment should validate the methodology and results of the ecosystem service assessment. The validation process should involve the commissioners. It should be carried out [*internally to its organisation; externally and involve third parties such as stakeholders; both internally to the contractor's organisation, and involved third parties such as stakeholders*]. The potential contractor should clearly describe the validation process that they expect to put in place during the assessment.

Back to the template landing page

3. VALIDATION

3.b. Should the results be compared to other published reference data in the validation process?

Answer yes, no, maybe, or don't know. Reference data are baseline or benchmark datasets or already published figures that will be used to compare the results of the ecosystem service assessment. Commissioners might want to list some key datasets or figures they would like the results to be comparable with. Alternatively, stakeholders can be consulted to identify these references.

Answer in this box (yes, no, maybe, or don't know).

Why is this question relevant to me?

Comparing results to other published data can give an estimation of the quality of the results. If they are expected to be close to some already published figures but are very different, it is an indication that the methodological approach chosen might have some flaws or can significantly improve the approach taken in previously published reports.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on uncertainty!

Give me an example answer

The results of the assessment can be compared with specific studies done on certain ecosystem services in parts of National Park Mainland.

Terms of Reference preformatted text

The validation process of the ecosystem service assessment should notably include a comparison of the results with other key published data including [*if relevant, list the key data and figures*]. The potential differences should be discussed and explained.

Back to the template landing page

4. STAKEHOLDER ENGAGEMENT

4.a. At which step(s) of the ecosystem service assessment should the stakeholders be informed, consulted, or participate?

Identify when the stakeholders should be informed, consulted or participate in the ecosystem service assessment. Stakeholders are any group, organisation or individual who are interested in and / or might be affected by ecosystem services and any change made to their flows. In the context of an ecosystem service assessment, stakeholders notably include those who directly or indirectly benefit from the service, those who try to regulate the use, the experts in the science of ecosystem services, or the data holders. For this, the commissioner can choose from the table below:

Possible steps of the assessment process at which involving stakeholders	Description	Stakeholder should be involved in this step
Development of the methodology	**Consult/participate:** stakeholders have a direct role in designing the approach. It might be relevant to involve stakeholders with key technical expertise to overcome some very specific technical difficulties in the methodological approach. **Inform:** provide updates and transparency on the progress of the assessment.	YES / NO / MAYBE / DON'T KNOW
Validation of the methodology (*e.g.,* metrics, indicators, data, choice of scenarios and analyses)	**Consult/participate:** stakeholders have an advisory role as the method has already been selected by the potential contractors and / or commissioners. It might be relevant to involve stakeholders with key technical expertise to quality ensure that the methodology that will be applied meets the minimum technical requirements for the objective(s) of the assessment. **Inform:** provide updates and transparency on the progress of the assessment.	YES / NO / MAYBE / DON'T KNOW
Validation of the results	**Consult/participate:** stakeholders with specific and different source of expertise may be a strong support for validating the results of the assessment. This gives more credibility and legitimacy to the results. It also ensures that they are understood by future potential users and compliant with their use(s). **Inform:** provide updates and transparency on the progress of the assessment.	YES / NO / MAYBE / DON'T KNOW
Data collection	**Consult/participate:** stakeholders can directly participate in data collection as respondents, and / or can help disseminate the means of collecting data such as surveys. This notably facilitates the inclusion of local knowledge in the assessment.	YES / NO / MAYBE / DON'T KNOW

	Inform: provide updates and transparency on the progress of the assessment.	
Interpretation of the results	**Consult/participate:** stakeholders have local and practical knowledge that can help contextualise the results, avoid misinterpretation and identify gaps. This also supports a sense of ownership of the results of the assessment by other stakeholders. **Inform:** provide updates and transparency on the progress of the assessment.	YES / NO / MAYBE / DON'T KNOW
Communication and dissemination of the results	**Consult/participate:** stakeholders can directly participate in the co-creation and dissemination of the key messages from the results of the assessment. Alternatively, they can also be used as channels of dissemination and communication of the results to a larger audience. **Inform:** support a communication that directly speaks to stakeholders, which in turn can help with a potential uptake of the results of the assessment.	YES / NO / MAYBE / DON'T KNOW
Other (specify)		YES / NO / MAYBE / DON'T KNOW

Why is this question relevant to me?

Stakeholder involvement provides legitimacy to the assessment process, it also ensures that results are communicated properly and that everyone impacted by the assessment is being included. Finally, it facilitates continuous feedback on methods, results and data used, which participates in the validation of the assessment.

Involving stakeholders can however be costly and it is important to do it strategically and identify the steps at which their involvement is necessary to choose the most adapted format of engagement.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on uncertainty and social benefits

Give me an example answer

Stakeholders participate in the validation of the results and communication of the results.

Terms of Reference preformatted text

A clear strategy should be put in place to inform, consult and facilitate stakeholders' participation at different stages of the assessment including [*list the steps of the assessment in which stakeholders should be informed, consulted or participate*].

4. STAKEHOLDER ENGAGEMENT

4.b. What methods should be used for involving stakeholders in the ecosystem service assessment?

Describe how the stakeholders should be involved. Stakeholders are any group, organisation or individual who are interested in and / or will or might be affected by ecosystem services and any change made to their flows. In the context of an ecosystem service assessment, stakeholders notably include those who directly or indirectly benefit from the service, those who try to regulate the use, the experts in the science of ecosystem services, or the data holders. For this, the commissioner can choose from the table below:

Possible ways of involving stakeholders during the assessment process	Description	Stakeholders should be involved this way
Newsletters or email updates	This format is very relevant to inform the stakeholders on the progress of the assessment.	YES / NO / MAYBE / DON'T KNOW
Websites	This format is very relevant to inform the stakeholders on the progress of the assessment, the events to come, or the communication and dissemination the results.	YES / NO / MAYBE / DON'T KNOW
Webinars	This format facilitates direct communication and interaction with stakeholders. It helps inform stakeholders on the progress of the assessment, the events to come, or communicating and disseminating results.	YES / NO / MAYBE / DON'T KNOW
Stakeholders events such as forums	This format brings stakeholders together, which can facilitate discussions between them and with the commissioners and / or contractors of the assessment. It helps with directly integrating the stakeholders and shows a higher level of consideration of their feedback and inputs.	YES / NO / MAYBE / DON'T KNOW
Advisory boards, expert groups, focus groups, or steering committees	These formats are direct channels of inputs and feedback from the stakeholders throughout the assessment process. It may be relevant to create an advisory group of stakeholders with specific expert or local knowledge to ensure the robust and safe development of the assessment.	YES / NO / MAYBE / DON'T KNOW
Public consultations	This format is very inclusive as it is not aimed at specific stakeholders only, but is open to all. Public consultations can take place in meetings, written forms, organised debates, deliberative groups etc. Note that Environmental Impact Assessment regulations may require certain procedure for public hearing.	YES / NO / MAYBE / DON'T KNOW
Workshops	This format allows a direct interaction with stakeholders and facilitates knowledge transfer but also the collection of local and expert knowledge relevant to the assessment.	YES / NO / MAYBE / DON'T KNOW
Other (specify)		YES / NO / MAYBE / DON'T KNOW

Back to the template landing page

Why is this question relevant to me?

Stakeholder involvement provides legitimacy to the assessment process, it also ensures that results are communicated properly and that everyone impacted by the assessment is being included. Finally, it facilitates continuous feedback on methods, results and data used, which participates in the validation of the assessment.

Involving stakeholders can however be costly and it is important to do it strategically and choose the most adapted format of engagement.

Where can I find more information or additional questions to detail my Terms of Reference?

Visit our advanced content on uncertainty and social benefits!

Give me an example answer

Stakeholders will be involved through advisory boards, expert groups, focus groups, or steering committees. Workshops should also be organised around the validation process. Public consultation is also a possibility on ecosystem services and other benefits that cannot be quantified.

Terms of Reference preformatted text

Stakeholder engagement should preferably include [*format(s) of stakeholder involvement chosen*] to [*reason(s) for which the format(s) has(have) been chosen*]

Back to the template landing page

Evaluation Criteria template

We recommend to read the 5.1 General Information prior to using the template

This template is different from the two previous ones for Frame & Scope and Methodology. Its aim is twofold:

i. **Support the writing of the Evaluation Criteria section of a Terms of Reference** by identifying scoping and methodological criteria that are essential to commissioners. For this, commissioners should use the blue column entitled "**WRITING EVALUATION CRITERIA**". At the end of subsection listing criteria, boxes with further information and preformatted Terms of Reference text are provided to facilitate the writing of the Evaluation Criteria section of a Terms of Reference.

ii. **Support the evaluation of proposals received** by commissioners once the call for tender has been closed. For this, commissioners should use the yellow column entitled "**EVALUATING PROPOSALS**", as well as the last two rows of each table of evaluation.

The guidance presented in this template is solely focused on the evaluation of the scope and methodology of a proposal, it does not provide any guidance on how to evaluate a proposal in its entirety. Refer to 5.1 General Information for a detailed description of use and limitations of this template.

Back to the main landing page

1. FRAME & SCOPE

	WRITING EVALUATION CRITERIA		EVALUATING PROPOSALS		
	Is this criteria essential to the assessment?		**Is this criteria included in the proposal received?**		
Criteria	YES	NO	YES	NO	DON'T KNOW
1.a. Include recommendations to decision-making					
1.b. Correspondence with the requested format of the results					
1.c. Include the ecosystem types of interest					
1.d. Include the ecosystem services of interest					
1.e. Include the correct the spatial extent(s)					
1.f. Include the correct time period and temporal scale for the results					
1.g. Include the correct spatial scales for the results					
1.h. Quantify ecosystem services in biophysical terms					
1.i. Quantify ecosystem services in terms of their economic value in monetary terms					
1.k. Quantify ecosystem service supply, use, demand					
1.l. Include an evaluation of sustainable use and / or supply					
1.m. Include social aspects					

	YES	NO	YES	NO	DON'T KNOW
1.n. Include health aspects					
1.o. Include an evaluation of ecosystem condition					
1.p. Include a stakeholder analysis					
Qualitative appreciation of the proposal					
	Qualitative score: **/5**				

Where can I find more information to add more criteria and / or evaluate proposals?
Visit our advanced content!

What are the associated Frame & Scope template sections?

Criteria	Frame & Scope section
1.a.	1. Context and Purpose
1.b.	2.3 Formats of the deliverables
1c.-1.d.	3.1 Ecosystem type(s) and service(s)
1.e.	3.2 Spatial characteristics
1.f.-1.g.	3.3 Temporal characteristics
1.h.-1.o.	3.4 Valuation of ecosystem services

What do these terms mean?

Spatial extent: refers to the geographical location and overall size of the area in which the ecosystem service assessment should be conducted. For more information refer to the Frame & Scope section above.

Time period: the interval of time that the assessment should cover, for example 2020-2024. For more information refer to the Frame & Scope section above.

Temporal scale: the unit of time in which the results should be compiled such as annual average, quarterly average, monthly average. For more information refer to the Frame & Scope section above.

<u>Spatial scale</u>: the spatial level at which the results should be compiled. In this guidance, three options were offered: fully spatially explicit, ecological scales (*e.g.,* deciduous *versus* coniferous forests), and administrative scales (*e.g.,* regions, counties, municipalities). For more information refer to the Frame & Scope section above.

<u>Ecosystem service supply</u>: The supply of ecosystem services is the measure in biophysical units of the contribution of ecosystems to the economy or society more generally (i.e., what quantity is being supplied?). It can be valued in terms of their economic value in monetary units (i.e., how much does this supply represent in the economy?). For more information refer to the Frame & Scope section above.

<u>Ecosystem service use</u>: The use of ecosystem services refers to the measure in biophysical units of the utilisation or consumption of the service by users or those who benefit from the services (i.e., what quantity is being used?). It can be valued in terms of their economic value in monetary units (i.e., how much does this use represent in the economy?). For more information refer to the Frame & Scope section above.

<u>Ecosystem service demand</u>: The demand for ecosystem services is the measure in biophysical units of the need for specific ecosystem service(s) by society, particular stakeholder groups or individuals (i.e., what quantity of an ecosystem service is needed?). It can be valued in terms of their economic value in monetary units (i.e., how much does this need represent in the economy?). For more information refer to the Frame & Scope section above.

<u>Sustainable use and supply</u>: Sustainable use of ecosystem services means that the services will be maintained and won't be used until depletion, but will be available for future generations. Sustainable supply of ecosystem services means that they will continue to be provided in the long term and be available for future generations. For more information refer to the Frame & Scope section above.

<u>Terms of Reference preformatted text</u>

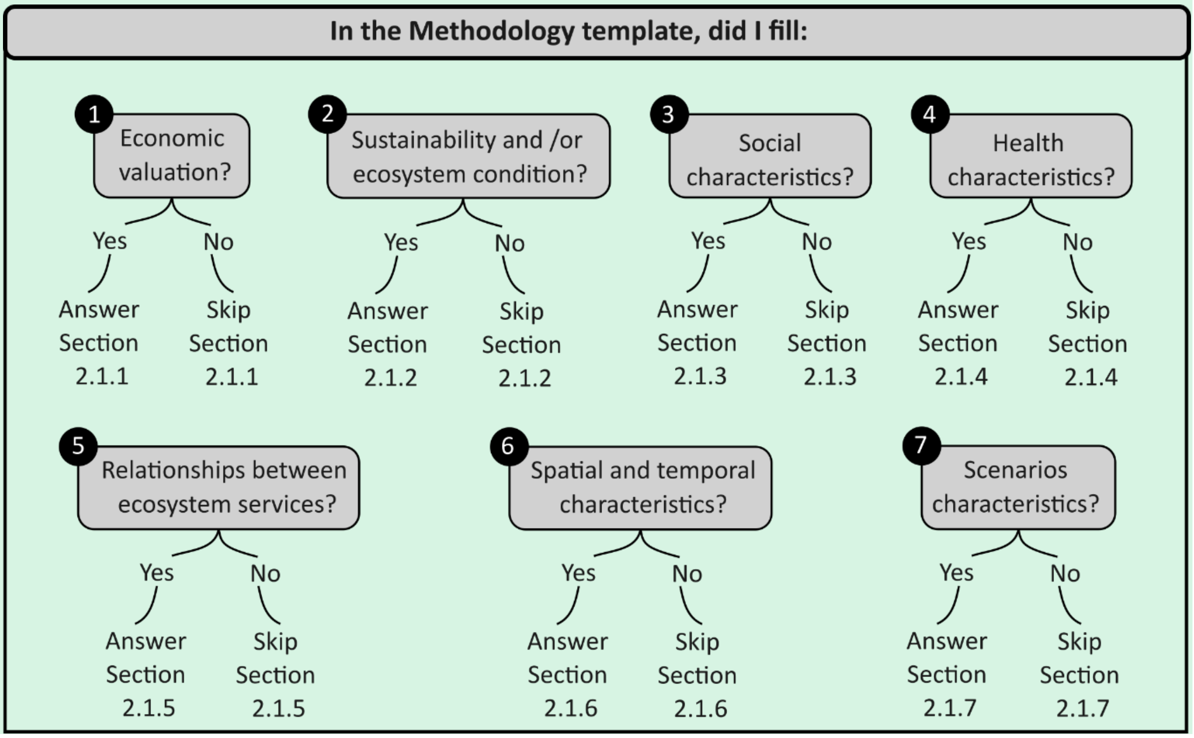
The proposals sent by the applicants will be evaluated according to their alignment with the frame & scope defined in the call, as well as the methodological approach(es) chosen to include the criteria of importance. Particular care should be given to:

- [*add the key components of the scope that are of high importance*].

Back to the template landing page

2. METHODOLOGY

2.1. Methodological characteristics of the outputs



	WRITING EVALUATION CRITERIA		**EVALUATING PROPOSALS**		
	Is this criteria essential to the assessment?		**Is this criteria included in the proposal received?**		
Criteria	YES	NO	YES	NO	DON'T KNOW
2.1.1. Economic valuation					
2.1.1.a. Specify and justify the economic valuation methods that will be used					
2.1.1.b. Quantify indicators of the contribution of ecosystem service(s) to economic development					
2.1.2. Sustainability and ecosystem condition					
2.1.2.a. Quantify the impact of ecosystem condition on ecosystem service supply					
2.1.2.b. Define and evaluate sustainable levels for the use and / or supply of ecosystem services					

Criteria	YES	NO	YES	NO	DON'T KNOW
2.1.2.c. Quantify relevant dimensions of ecosystem condition					
2.1.3. Social characteristics					
2.1.3.a. Disaggregate results of the ecosystem service assessment by social groups to identify beneficiaries					
2.1.3.b. Assess and report the attitudes, perceptions, and values of social groups towards ecosystem services					
2.1.4. Health characteristics					
2.1.4.a. Measure of health risks					
2.1.4.b. Measure of health benefits					
2.1.4.c. Measure of health outcomes					
2.1.4.d. Assess and report the attitudes, perceptions, and values of social groups towards health-relevant ecosystem services					
2.1.4.e. Establish health links with specific ecosystem types and / or ecosystem services					
2.1.5 Relationships between ecosystem services					
2.1.4.a. Identify potential interrelationships between ecosystem services and the benefits they provide					
2.1.6 Spatial and temporal characteristics					
2.1.6.a. Evaluate the spatial distribution of ecosystem service(s) and / or benefits					
2.1.6.b. Evaluate the temporal distribution of ecosystem service(s) and / or benefits					

Criteria	YES	NO	YES	NO	DON'T KNOW
2.1.7. Scenarios					
2.1.7.a. Include relevant scenarios					
Qualitative appreciation of the proposal					
	Qualitative score: **/5**				

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What are the associated Methodology template sections?

Criteria	Methodology section
2.1.1.a -2.1.1.b	1.1 Economic valuation
2.1.2.a -2.1.2.c	1.2 Sustainability and ecosystem condition
2.1.3.a – 2.1.3.b	1.3 Social characteristics
2.1.4.a – 2.1.3.e	1.4 Health characteristics
2.1.5.a	1.5 Relationships between ecosystem services
2.1.6.a – 2.1.6.b	1.6 Spatial and temporal characteristics
2.1.7.a	1.7 Scenarios characteristics

What do these terms mean?

Ecosystem condition: the quality of an ecosystem. For more information refer to the Methodology section above.

Sustainable levels: refers to the maximum consumption and provision of ecosystem services above which the ecosystem condition and ability to deliver services for future generations is maintained. For more information refer to the Methodology section above.

Health risks: refers to potential threats to good health, including drivers of disease emergence. For more information refer to the Methodology section above.

Health benefits: benefits includes goods or services which promote good health or offer avenues for positive health interventions. For more information refer to the Methodology section above.

Health outcomes: refers to the status of health as a result of an intervention or exposure to one or more conditions. For more information refer to the Methodology section above.

Interrelationships: refer to how different services or benefits influence each other. This influence can be either positively (synergies), negatively (trade-offs), or through dependence (interdependencies). For more information refer to the Methodology section above.

Scenarios: in modelling, future scenarios represent a range of potential futures that one would like to predict. For more information refer to the Methodology section above.

Terms of Reference preformatted text

The proposals sent by the applicants will be evaluated according to their alignment with the methodological characteristics defined in the call. The proposed methodological approach(es) to measure the methodological characteristics defined above including modelling methods, metrics and indicators should be clearly described and justify. Of particular importance are the following methodological characteristics:

- [*if relevant, key components of the methodological characteristics that are of high importance*];
- [*…*]

Back to the template landing page

2. METHODOLOGY

2.2. Uncertainty

	WRITING EVALUATION CRITERIA		EVALUATING PROPOSALS		
	Is this criteria essential to the assessment?		Is this criteria included in the proposal received?		
Criteria	YES	NO	YES	NO	DON'T KNOW
2.2.a. Assess and document uncertainties arising in the ecosystem service assessment					
2.2.b. Include a clear reporting format for uncertainties					
Qualitative appreciation of the proposal					
	Qualitative score: **/5**				

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What are the associated Methodology template sections?

Criteria	Methodology section
2.2.a. -2.2.b.	2. Uncertainty

What do these terms mean?

Uncertainty: Uncertainty refers to the degree of confidence we can have in the results of an assessment. It includes both natural variability and gaps in knowledge or data. Quantifying uncertainties is important as these can have implications for decision-making. For more information refer to the Methodology section above.

Terms of Reference preformatted text

The proposals sent by the applicants will be evaluated according to the following criteria:

- the approach taken to document and report uncertainties in relation to decision-making, modelling and scenarios if applicable.

Back to the template landing page

2. METHODOLOGY

2.3. Validation

	WRITING EVALUATION CRITERIA		EVALUATING PROPOSALS		
	Is this criteria essential to the assessment?		Is this criteria included in the proposal received?		
Criteria	YES	NO	YES	NO	DON'T KNOW
2.3.a. Include a clear workplan and timeline for the validation of the methodology, and / or results of the ecosystem service assessment					
2.3.b. Involvement of stakeholders in the validation process of the ecosystem service assessment					
2.3.c. Validation of the results of the ecosystem service assessment includes comparison with published reference data					
Qualitative appreciation of the proposal					
	Qualitative score: /5				

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What are the associated Methodology template sections?

Criteria	Methodology section
2. 3.a. -2.3.c.	3. Validation

What do these terms mean?

Validation: means evaluating the quality of the assessment including the scope, the methodology employed and the results. For more information refer to the Methodology section above.

Stakeholders: Any group, organisation or individual who are interested in and / or will or might be affected by ecosystem services and any change made to their flows. In the context of an ecosystem service assessment, stakeholders notably include those who directly or indirectly benefit from the service, those who try to regulate the use, the experts in the science of ecosystem services, or the data holders.

Published reference data: baseline or benchmark datasets or already published figures that will be used to compare the results of the ecosystem service assessment. For more information refer to the Methodology section above.

Terms of Reference preformatted text

The proposals sent by the applicants will be evaluated according to the following criteria:

- the approach taken to validate the methodology and results of the ecosystems service assessment.

Back to the template landing page

2. METHODOLOGY

2.4. Stakeholder engagement

	WRITING EVALUATION CRITERIA		EVALUATING PROPOSALS		
	Is this criteria essential to the assessment?		Is this criteria included in the proposal received?		
Criteria	YES	NO	YES	NO	DON'T KNOW
2.4.a. Include a clear workplan and timeline to inform, consult and / or make stakeholders participate in the ecosystem service assessment					
2.4.b. Propose relevant modes of stakeholder involvement throughout the ecosystem service assessment					
Qualitative appreciation of the proposal					
	Qualitative score: **/5**				

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What are the associated Methodology template sections?

Criteria	Methodology section
2.4.a.-2.4.b.	4. Stakeholder engagement

Stakeholders: Any group, organisation or individual who are interested in and / or might be affected by ecosystem services and any change made to their flows. In the context of an ecosystem service assessment, stakeholders notably include those who directly or indirectly benefit from the service, those who try to regulate the use, the experts in the science of ecosystem services, or the data holders.

Terms of Reference preformatted text

The proposals sent by the applicants will be evaluated according to the following criteria:

- the approach taken and timeline to inform, consult and involve stakeholders throughout the ecosystem service assessment. The applicants will identify the foreseen ways of engaging with the stakeholders throughout the assessment. The applicants might also refer to specifically relevant stakeholders in alignment with the scope of the assessment.

Back to the template landing page

SELINA

SCIENCE FOR EVIDENCE-BASED AND SUSTAINABLE DECISIONS ABOUT NATURAL CAPITAL

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Table A.1 - Primary valuation methods, applicability to ecosystem services, examples and limitations. *ES stands for ecosystem service. Source: (adapted from Table A2, Brander 2013)*

Valuation method	Approach	Data requirements and sources	Application to ecosystem services	Example ecosystem service	Limitations
Market prices	Prices for ES that are directly observed in markets.	Prices of some ES can be obtained from markets or surveys of businesses and households.	ES that are traded directly in markets.	Timber and fuel wood from mangroves; fish; ecotourism.	Market prices can be distorted e.g. by subsidies. Most ES are not traded in markets.
Public pricing	Public expenditure or monetary incentives (taxes/subsidies) for ES as an indicator of value.	Data on public expenditures on the provision of ES obtained from government reports or key informants.	ES for which there are public expenditures.	Watershed protection to provide drinking water; Purchase of land for protected area.	No direct link to preferences of beneficiaries.
Defensive expenditure	Expenditure on protection of ES.	Data on public or private expenditure obtained from government reports, key informants, or surveys of businesses and households.	ES for which there is public or private expenditure for its protection.	Recreation and aesthetic values from protected areas.	Only applicable where direct expenditures are made for environmental protection related to provision on an ES. Provides lower bound estimate of ES benefit.
Replacement cost	Estimate the cost of replacing an ES with a man-made service.	Estimates of construction costs can be obtained from experts or based on past investments.	ES that have man-made equivalents.	Coastal protection by dunes (replaced by seawalls); water storage and filtration by wetlands (replaced by reservation and filtration plant).	No direct relation to ES benefits. Over-estimates value if society is not prepared to pay for man-made replacement. Under-estimates value if man-made replacement does not provide all of the benefits of the original ecosystem.

Valuation method	Approach	Data requirements and sources	Application to ecosystem services	Example ecosystem service	Limitations
Restoration cost	Estimate cost of restoring degraded ecosystems to ensure provision of ES.	Estimates of restoration costs can be obtained from experts or based on past investments.	Any ES that can be provided by restored ecosystems.	Coastal protection by dunes; water storage and filtration by wetlands.	No direct relation to ES benefits. Over-estimates value if society is not prepared to pay for restoration. Under-estimates value if restoration does not provide all of the benefits of the original ecosystem.
Damage cost avoided	Estimate damage avoided due to ecosystem service.	Data on past damage costs and frequencies can be obtained from government reports and household surveys.	Ecosystems that provide storm, flood or landslide protection to houses or other assets.	Coastal protection by dunes, mangroves and reefs; river flow control by wetlands; landslide protection by forests.	Difficult to quantify changes in risk of damage to changes in ecosystem condition.
Social cost of carbon (SCC)	The monetary value of damages caused by emitting one tonne of CO_2 in a given year. The social cost of carbon (SCC) therefore also represents the value of damages avoided for a one tonne reduction in emissions.	Estimates of the SCC can be obtained from Integrated Assessment Models of climate-economy impacts and published summaries of model results.	Carbon storage and sequestration.	Carbon sequestered and stored by protected or restored mangroves.	SCC is a specific application of the "damage cost avoided" method. SCC is characterized by high modelling uncertainties and partial coverage of climate change impacts.

Valuation method	Approach	Data requirements and sources	Application to ecosystem services	Example ecosystem service	Limitations
Opportunity cost	The next highest valued use of the resources used to produce an ecosystem service.	Data on the value of alternative land uses (e.g. agriculture, aquaculture, housing, hotels etc.) can be obtained from markets and surveys of businesses and households.	All ecosystem services.	The opportunity cost of ecosystem services from a natural ecosystem might be the value of agricultural output if the land is converted to agriculture instead of conserved in a natural state.	Measures the cost of providing ecosystem services instead of the benefit.
Net factor income (residual value)	Revenue from sales of a marketed good with an ES input minus the cost of other inputs.	Revenues can be obtained from markets; costs can be obtained from business surveys.	Ecosystems that provide an input in the production of a marketed good.	Filtration of water by wetlands; commercial fisheries supported by mangroves and reefs.	Tendency to over-estimate values since all normal profit is attributed to the ES.
Production function	Statistical estimation of production function for a marketed good with an ES input.	Data on production, inputs, costs and revenues can be obtained from business surveys.	Ecosystems that provide an input in the production of a marketed good.	Soil quality or water quality as an input to agricultural production.	Technically difficult. High data requirements.

Valuation method	Approach	Data requirements and sources	Application to ecosystem services	Example ecosystem service	Limitations
Input-Output Models	Quantifies the interdependencies between economic sectors in order to measure the impacts of changes in one sector to other sectors in the economy. Ecosystems can be incorporated as distinct sectors.	Data on production inputs, outputs and prices for multiple economic sectors can be obtained from government statistics. Data on ecosystem inputs and outputs can be observed or modelled using bio-physical methods.	Ecosystem services with direct and indirect use values, particularly inputs into production.	Ecosystem inputs into agriculture; or into the tourism sector.	Requires substantial data on ecosystem-economy linkages to parameterise connections between sectors.
Hedonic pricing	Estimate influence of environmental characteristics on price of marketed goods (usually residential property).	Data on house prices and characteristics can be obtained from estate agents or public records. Data on environmental characteristics can be observed or modelled using bio-physical methods.	Environmental characteristics that vary across goods (usually houses).	Urban green open space; air quality moderated by ecosystems.	Technically difficult. High data requirements. Limited to ES that are spatially related to property locations.
Travel cost	Estimate demand for ecosystem recreation sites using data on travel costs and visit rates.	Data on travel costs and visit rates can be obtained through visitor surveys.	Recreational use of ecosystems.	Recreational use of beaches, reefs, national parks etc.	Technically difficult. High data requirements. Limited to valuation of recreation. Complicated for trips with multiple purposes or to multiple sites.

Valuation method	Approach	Data requirements and sources	Application to ecosystem services	Example ecosystem service	Limitations
Contingent valuation	Ask people to state their WTP for an ES through surveys.	Data collected through public surveys.	All ecosystem services.	Biodiversity; recreation; landscape aesthetics; flood risk attenuation.	Expensive and technically difficult to implement. Risk of biases in design and analysis.
Choice modelling (choice experiment)	Ask people to make trade-offs between ES and other goods to elicit WTP.	Data collected through public surveys.	All ecosystem services.	Biodiversity; recreation; landscape aesthetics; flood risk attenuation.	Expensive and technically difficult to implement. Risk of biases in design and analysis.
Group / participatory valuation	Ask groups of stakeholders to state their WTP for an ES through group discussion.	Data collected in workshop settings.	All ecosystem services.	Biodiversity; recreation; landscape aesthetics; flood risk attenuation.	Risk of biases due to group dynamics.

3. Value transfer methods

Decision-making often requires information quickly and at low cost. New 'primary' valuation research, however, is generally time consuming and expensive. For this reason, there is interest in using information from existing primary valuation studies to inform decisions regarding impacts on ecosystems that are of current interest. This transfer of value information from one context to another is called "value" or "benefit transfer".

Value transfer is the use of research results from existing primary studies at one or more sites or policy contexts ("study sites") to predict welfare estimates or related information for other sites or policy contexts ("policy sites"). Value transfer is also known as benefit transfer but since the values that are transferred may be costs as well as benefits, the term value transfer is more generally applicable.

In addition to the need for expeditious and inexpensive information, there is often a need for information on the value of ecosystem services at a different geographic scale from that at which primary valuation studies have been conducted. So even in cases where some primary valuation research is available for the ecosystem of interest, it is often necessary to extrapolate or scale-up this information to a larger area or to multiple ecosystems in the region or country. Primary valuation studies tend to be conducted for specific ecosystems at a local scale whereas the information required for decision-making is often needed at a regional, national or multi-national scale. Value transfer therefore provides a means to obtain information for the scale that is required.

The number of primary studies on the value of ecosystem services is substantial and growing rapidly. This means that there is a growing body of evidence to draw on for the purposes of transferring values to inform decision-making. With an expanding information base, the potential for using value transfer is improved.

Value transfer can potentially be used to estimate values for any ecosystem service, provided that there are primary valuations of that ecosystem service from which to transfer values. Value transfer methods have been employed widely in national and global ecosystem assessments, value mapping applications and policy appraisals. The use of value transfer is widespread but requires careful application. The alternative methods of conducting value transfer are described here:

- **Unit value transfer** uses values for ecosystem services at a study site, expressed as a value per unit (usually per unit of area or per beneficiary), combined with information on the quantity of units at the policy site to estimate policy site values. Unit values from the study site are multiplied by the number of units at the policy site. Unit values can be adjusted to reflect differences between the study and policy sites (*e.g.,* income and price levels).

- **Value function transfer** uses a value function estimated for an individual study site in conjunction with information on parameter values for the policy site to calculate the value of an ecosystem service at the policy site. A value function is an equation that relates the value of an ecosystem service to the characteristics of the ecosystem and the beneficiaries of the ecosystem service. Value functions can be estimated from a number of primary valuation methods including hedonic pricing, travel cost, production function, contingent valuation and choice experiments.

- **Meta-analytic function transfer** uses a value function (see above) estimated from the results of multiple primary studies representing multiple study sites in conjunction with information on parameter values for the policy site to calculate the value of an ecosystem service at the policy site. Since the value function is estimated from the results of multiple studies it is able to represent and control for greater variation in the characteristics of ecosystems, beneficiaries and other contextual characteristics. This feature of meta-analytic function transfer provides a means to account for simultaneous changes in the stock of ecosystems when estimating economic values for ecosystem services (i.e., the "scaling up problem"). By including an explanatory variable in the data describing each "study site" that measures the scarcity of other ecosystems in the vicinity of the "study site", it is possible to estimate a quantified relationship between scarcity and ecosystem service value. This parameter can then be used to account for changes in ecosystem scarcity when conducting value transfers at large geographic scales.

These three principal methods for transferring ecosystem service values are summarized in **Table A.2** together with their respective strengths and weaknesses. The choice of which value transfer method to use to provide information for a specific policy context is largely dependent on the availability of primary valuation estimates and the degree of similarity between the study and policy sites. In cases where value information is available for a highly similar study site, unit value transfer may provide the most straightforward and reliable means of conducting value transfer. On the other hand, when study sites and policy sites are different, value function or meta-analytic function transfer offers a means to systematically adjust transferred values to reflect those differences. Similarly, in the case that value information is required for multiple different policy sites, value function or meta-analytic function transfer may be a more accurate and practical means for transferring values. Using meta-analytic functions that include a parameter for ecosystem scarcity provides a means to account for simultaneous changes in the stock of ecosystem on the value of all ecosystem services (*i.e.,* more accurately "scale-up" ecosystem service values).

Table A.2 - Value transfer methods, strengths, weaknesses. *ES stands for ecosystem service. Source: adapted from Table 3, Brander (2013)*

Method	Approach	Strengths	Weaknesses
Unit value transfer	Select appropriate values from existing primary valuation studies for similar ecosystems and socio-economic contexts. Adjust unit values to reflect differences between study and policy sites (usually for income and price levels).	Simple	Unlikely to be able to account for all factors that determine differences in values between study and policy sites. Value information for highly similar sites is rarely available.
Value function transfer	Use a value function derived from a primary valuation study to estimate ES values at policy site(s).	Allows differences between study and policy sites to be controlled for (e.g. differences in population characteristics).	Requires detailed information on the characteristics of policy site(s).
Meta-analytic function transfer	Use a value function estimated from the results of multiple primary studies to estimate ES values at policy site(s).	Allows differences between study and policy sites to be controlled for (e.g. differences in population characteristics, area of ecosystem, abundance of substitutes etc.). Practical for consistently valuing large numbers of policy sites.	Requires detailed information on the characteristics of policy site(s). Analytically complex.